

Color Journey Engine: A Perceptually-Uniform Palette Generation Specification

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Abstract

This paper specifies the Color Journey Engine, a deterministic algorithm for generating perceptually-uniform color palettes. Built on the OKLab color space, the engine uses cubic Bézier curves to construct “journeys” through perceptual color space, producing coherent palettes from one to five anchor colors. A novel single-anchor expansion algorithm—to our knowledge not previously formalised—generates harmonious variations from a single color by exploiting lightness-weighted directions in perceptual space. The specification defines perceptual constraints based on just-noticeable difference thresholds, user-facing style controls for temperature, intensity, and smoothness, and multiple loop strategies for extended sequences. A two-layer gamut management approach ensures valid sRGB output while preserving design intent. The engine operates as a pure function with complete determinism, achieving 5.6 million colors per second in the reference implementation. This specification serves as both a formal definition and implementation guide for reproducible palette generation in user interfaces, data visualization, and generative art applications.

Keywords: color palette generation, OKLab, perceptual uniformity, Bézier curves, gamut mapping, deterministic algorithms

Note on Spelling: This specification uses American English spelling conventions (“color” not “colour”, “neighbor” not “neighbour”, etc.) for consistency with international technical standards, API naming conventions (e.g., CSS Color Level 4, web standards), and the broader computer science and software engineering literature. This choice reflects the engine’s intended use in software development contexts where American spelling is the de facto standard for programming interfaces and technical documentation.

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Chapter 1

Introduction and Scope

1.1 Motivation

Every domain working with color—UI design, data visualisation, generative art—requires coherent palettes. Yet producing them is deceptively hard. Ad-hoc methods (RGB interpolation, color wheels) fail fundamentally: adjacent colors feel inconsistent, some nearly identical while others jump dramatically. This is not a cosmetic problem, it breaks user experience and data clarity.

The fundamental problem is that commonly-used color spaces (sRGB, HSL, HSV) are not *perceptually uniform*. Equal numerical steps do not correspond to equal perceived differences. A mathematically elegant gradient in RGB may appear uneven to the human eye, with certain hue regions changing faster than others.

This specification addresses the need for a *principled, reproducible* approach to palette generation—one grounded in perceptual color science, where mathematical operations correspond meaningfully to visual experience.

Audience and Purpose

This document serves as an internal technical reference for the engineering team developing and maintaining the C-core Color Journey Engine implementation. It documents design rationale, mathematical foundations, and architectural decisions to enable team alignment—ensuring we can say “see §3.3” and be immediately on the same page.

The paper prioritises the “why” over the “how”: implementation details are covered in separate technical documentation, while this specification captures the reasoning behind design choices, the perceptual science foundations, and the constraints that shape the system.

1.2 Scope Definition

The Color Journey Engine produces **ordered sequences of discrete color swatches** by constructing perceptually-aware paths through color space. Given one to five anchor colors and a set of style parameters, the engine generates N swatches such that adjacent colors are perceptually distinguishable yet smoothly related.

What the Engine Does

- Generates deterministic color palettes using perceptual color science
- Outputs discrete swatch arrays (not continuous functions)
- Enforces perceptual constraints ensuring distinguishability
- Provides diagnostic information about palette quality

What the Engine Does Not Do

The engine deliberately excludes capabilities that belong to callers or separate systems:

- **Accessibility checking**—WCAG contrast verification is caller responsibility
- **Animation control**—temporal playback logic belongs to the application
- **Color naming**—a separate knowledge domain
- **Color management**—ICC profiles and device calibration are external concerns
- **Continuous curve evaluation**—the curve is an internal construction device, not an exposed API

1.3 Design Principles

Five core principles guide the engine’s design:

Perceptual Uniformity as Foundation. All design decisions rest on the perceptual uniformity of OKLab color space Ottosson, 2020. Mathematical distances correspond to perceived differences, enabling meaningful constraints and smooth transitions.

Discrete Output, Continuous Thinking. The engine constructs continuous paths through color space internally but outputs discrete swatches. The “journey” is a construction metaphor—users receive a finite list of colors, not an interpolation function.

Constructive, Not Prescriptive. Colors emerge *as if* travelling a perceptual path, providing coherence and natural ordering. Users may apply the resulting swatches in any manner; the ordering is a construction device, not a usage constraint.

Presets Encode Expertise. Presets capture expert knowledge about which parameter combinations achieve specific aesthetic goals—they are not simplifications for novices. A “Cinematic” preset encodes design wisdom about dramatic color arcs.

Graceful Degradation. Rather than failing on unusual inputs, the engine adapts. Grey anchors expand along the lightness axis; extreme colors adjust their expansion direction; pathological inputs produce documented fallback behavior.

1.4 Positioning and Novel Contributions

Prior Art: Three Paradigms of Palette Generation

Palette generation approaches fall into three broad paradigms, each with distinct strengths and limitations.

Geometric Harmony Rules. Traditional *color harmony rules*—complementary, triadic, analogous, split-complementary—define geometric relationships on the color wheel (Itten, 1961). These rules have aesthetic merit and deep historical roots in art education, but they operate in perceptually non-uniform spaces (typically HSL or HSV) and provide no guarantees about perceived differences between generated colors. Two colors deemed “complementary” by wheel position may or may not be perceptually distinguishable depending on their lightness and chroma values.

Interpolation-Based Methods. *Interpolation-based approaches* generate colors by blending between endpoints. Most creative tools (CSS gradients, design software) use linear interpolation in RGB or HSL space, producing results that can appear perceptually uneven—muddy midpoints, abrupt hue shifts through grey regions, and inconsistent step sizes (Stone, Szafr, and Setlur, 2014). Even when tools support perceptual spaces, they typically output continuous gradients rather than discrete palettes with distinguishability guarantees (Madsen, 2017).

Data-Driven Methods. More recent academic work has explored *data-driven approaches* that learn palette patterns from images or existing designs. Liu et al. (Liu

et al., 2013) demonstrate image-driven harmonious color extraction using saliency-hue clustering with harmony optimisation. Such methods excel at capturing aesthetic patterns from existing content but require training data, produce less predictable outputs, and are difficult to parameterise for specific design goals. The Color Journey Engine takes a deliberately different path: deterministic construction with explicit, interpretable parameters.

The Shift to Perceptual Color Spaces

Modern tools increasingly adopt perceptually uniform color spaces to address the limitations of geometric and RGB-based methods. CIELAB CIE, 1976 was the first widely-used perceptual space but exhibits known uniformity issues, particularly in blue-violet regions Mahy, Van Eycken, and Oosterlinck, 1994. CIEDE2000 Luo, Cui, and Rigg, 2001 improved color difference calculation but remains computationally complex.

OKLab, introduced by Björn Ottosson in 2020 Ottosson, 2020, represents a significant advance: CAM16-level perceptual uniformity Safdar et al., 2017 with computational efficiency suitable for real-time applications. Independent analysis has validated OKLab’s perceptual uniformity claims for gradient and color manipulation applications Levien, 2021. OKLab has achieved rapid industry adoption, appearing in CSS Color Level 4 W3C CSS Working Group, 2022, Adobe Photoshop (as the default gradient interpolation space), Unity, and Godot. The Color Journey Engine builds on this established foundation.

Our Contribution

The Color Journey Engine introduces a *novel combination* of established techniques:

- **Continuous Bézier paths** through OKLab space Farin, 2002, providing flexible curve shapes with well-understood mathematical properties
- **Arc-length parameterisation** Piegl and Tiller, 1997; Kamermans, 2023 for perceptually uniform sampling along the constructed path
- **Perceptual constraints** ($\Delta_{\min}$, $\Delta_{\max}$) ensuring distinguishability without jarring jumps, grounded in just-noticeable difference research
- **Single-anchor mood expansion**—to our knowledge, a novel approach—using lightness-direction heuristics to generate coherent palettes from one color
- **Two-layer gamut management** Morović, 2008 preventing and correcting out-of-gamut colors while preserving hue, aligned with CSS Color Level 4 practices Verou

and Lilley, [2024](#)

Our contribution lies not in the individual techniques—OKLab, Bézier curves, arc-length sampling, and gamut mapping are established—but in their integration into a coherent system with explicit perceptual constraints and a deterministic, reproducible API. Unlike data-driven methods that require training corpora, the Color Journey Engine produces predictable, parameterisable results. Unlike simple interpolation, it guarantees perceptual distinguishability. Unlike geometric harmony rules, it operates in perceptually uniform space with explicit distance constraints.

The resulting implementation achieves 5.6 million colors per second, enabling real-time palette generation for interactive applications.

1.5 Paper Organization

This specification proceeds as follows:

- [§2.1](#): Perceptual foundations—Riemannian structure, OKLab space, and temporal perception
- [§3.1–3.6](#): Journey construction—anchors, Bézier curves, and sampling
- [§4.1–4.5](#): Perceptual constraint framework
- [§5.1–6.4](#): Style controls and modes of operation
- [§7.1–8.4](#): Loop strategies and gamut management
- [§9.1–11.4](#): Determinism, API design, and caller responsibilities
- [§12.1–12.3](#): Summary and future directions

Appendices provide preset reference tables, mathematical notation, implementation examples, and a quick-reference guide.

Chapter 2

Perceptual Color Foundations

2.1 Perceptual Foundations

The design of temporally coherent color progressions rests on a rigorous foundation in perceptual color science. This section establishes three principles that underpin subsequent design decisions: (1) perceptual color difference is well modeled by a Riemannian metric rather than a globally Euclidean geometry; (2) no single global three-dimensional Euclidean coordinate system can be perfectly perceptually uniform without introducing distortion; and (3) temporal color perception differs materially from spatial color perception. These are empirically supported constraints with direct implications for how color journeys should be constructed.

2.2 Color Space as Riemannian Manifold

2.2.1 Theoretical Framework

Recent comprehensive measurements of human color discrimination thresholds provide strong evidence that perceptual color difference is well modeled as a *Riemannian manifold*, a space that is locally Euclidean but globally curved (Hong et al., 2024). This has direct implications for color interpolation and distance measurement.

In a Euclidean space, the shortest path between two points is a straight line, and distances can be computed using Pythagorean geometry. In a Riemannian manifold, the shortest path is a *geodesic*, a curve that follows the intrinsic geometry of the space (Hong et al., 2024). Perceptual difference can be usefully approximated by the geodesic distance implied by the local discrimination metric, particularly in the small-step regime relevant to smooth transitions (Hong et al., 2024).

Hong et al. (2024) characterized this structure empirically through approximately

6,000 discrimination threshold trials per participant, mapping elliptical contours of equal perceptual distance across color space.¹

The metric tensor

In practice, discrimination varies with location in color space. This can be modeled by a position-dependent *metric tensor* $g(\mathbf{x})$ that defines local perceptual distance via the line element:

$$ds^2 = d\mathbf{x}^\top g(\mathbf{x}) d\mathbf{x} \quad (2.1)$$

where $\mathbf{x}$ represents coordinates in color space and ds is the local perceptual distance. Distances along specific directions (e.g., hue at fixed lightness and chroma) follow from the corresponding components of g integrated along a path. Hong et al.’s findings reveal systematic patterns in the metric tensor components:

1. **Radial orientation:** Major axes of threshold ellipses are consistently oriented toward the achromatic center (Hong et al., 2024)
2. **Lightness compression:** Thresholds are smaller along the lightness dimension than along chromatic axes
3. **Geodesic curvature:** Perceptually optimal paths between distant colors are curved, not straight, in standard colorimetric coordinates

This Riemannian framework has historical foundations extending to Schrödinger’s extension of Helmholtz’s line element (Roberti and Peruzzi, 2023).² Roberti and Peruzzi (2023) note that Schrödinger distinguished between *lower color metrics* (where affine geometry suffices for color mixture) and *higher color metrics* (where Riemannian geometry is necessary to translate color differences into perceptual distances). The engine’s goals align with the domain of higher color metrics.

2.2.2 Practical Implications

For color journey construction, the Riemannian structure has three critical consequences:

¹Hong et al.’s study employed the Bayesian Wishart Process Psychophysical Model (WPPM) within a Riemannian manifold framework, providing comprehensive characterization of the isoluminant plane through extensive empirical measurement.

²Roberti and Peruzzi provide a comprehensive analysis of Schrödinger’s (1920) Riemannian color metric and its improvements over Helmholtz’s original formulation—critical foundations for understanding perceptual color space geometry and geodesic paths in color applications.

Linear interpolation in RGB is perceptually nonuniform Naïve component-wise interpolation in gamma-encoded sRGB between saturated blue (0, 0, 255) and saturated yellow (255, 255, 0) produces a midpoint near (127, 127, 127), a desaturated grey, because the straight line passes through the achromatic region. The perceptually smooth path would curve around the exterior of color space.

Distance measurements require appropriate metrics A ΔE of 10 in a highly saturated region may be perceptually equivalent to a ΔE of 5 in a near-neutral region if the underlying metric tensor differs. Hong et al.’s ellipse measurements (Hong et al., 2024) provide the data needed to construct position-dependent distance corrections.

Gamut mapping must follow geodesics When projecting out-of-gamut colors to displayable coordinates, the projection should follow perceptual geodesics rather than straight lines in RGB (§8.3). Hong et al.’s finding that discrimination ellipses orient radially toward the achromatic center (Hong et al., 2024) justifies reducing saturation (moving toward achromatic) as the perceptually minimal modification.

2.2.3 Computational Approximation

True geodesic computation in a Riemannian manifold requires solving differential equations or variational problems, which is computationally expensive for real-time applications. The engine adopts a pragmatic approach: work in a color space where *Euclidean straight lines approximate geodesics sufficiently well*. This is the definition of perceptual uniformity.

Caveat on global structure and suprathreshold perception

While a Riemannian metric is an effective model of *local* discrimination structure, evidence suggests that suprathreshold dissimilarity judgements may be nonadditive (for example exhibiting diminishing returns for large separations) (Bujack et al., 2022).³ Accordingly, we treat metric-based distance as a practical local approximation for constructing smooth paths with bounded per-step changes, and we present global distance modelling as an open area for further empirical refinement.

Modern perceptually uniform color spaces (OKLab, CAM16-UCS) are designed such that Euclidean distance ΔE in the coordinate representation approximates geodesic distance in the underlying perceptual manifold (Ottosson, 2020; Safdar et al., 2017). The

³Bujack et al. argue that many perceptual color metrics are valid primarily for small differences, with large color differences exhibiting nonadditive behavior under psychophysical testing—a departure from strict Riemannian geometry.

approximation is not perfect, local curvature remains (Hong et al., 2024), but the error is small enough for practical interpolation tasks.

The engine thus achieves a balance: acknowledging the Riemannian nature of color space theoretically while exploiting Euclidean approximations computationally. All subsequent constraints and optimization objectives ($\Delta_{\min}$, $\Delta_{\max}$, velocity weights) are interpreted as *locally Euclidean* values in OKLab units and are intended to operate in the small-step regime.

2.3 The 720° Topology and Euclidean Impossibility

2.3.1 Mathematical Statement

A fundamental result in perceptual color theory is that no single global 3D Euclidean coordinate system can make perceptual distance everywhere equal to Euclidean distance without distortion (Nölle, Suda, and Boxleitner, 2013, p. 2). Consequently, practical 3D color spaces must either accept nonuniformity or treat distance as position-dependent (via a metric tensor). This is not merely a limitation of current measurement techniques; it reflects a mathematical constraint closely related to the “super-importance of hue” first identified by Judd (Judd, 1940, p. 11).⁴

Nölle et al. (2013) provide a derivation within their proposed H2SI geometry.⁵ They define a saturation-dependent circumference term $s_2(S)$ for a constant-lightness, constant-saturation hue circle as:

$$s_2(S) = \sqrt{\frac{5}{2}} S \int_0^{2\pi} dH = \pi\sqrt{10} S \quad (2.2)$$

where S denotes saturation in the H2SI construction and $s_2(S)$ is the circumference term (in the authors’ metric units) (Nölle, Suda, and Boxleitner, 2013, p. 9, Eq. (3.10)). This expression is the H2SI specialization of the more general hue-arc-length integral implied by Eq. 2.1, i.e., integrating the appropriate metric component along a constant- L , constant- S hue path. They then compare this circumference to the corresponding radial distance from a saturated point toward the neutral axis, yielding the ratio (Nölle, Suda, and Boxleitner, 2013, p. 9, Eq. (3.11)):

⁴Judd’s concept of the “super-importance of hue” established that hue differences contribute disproportionately to perceived color difference in ways that cannot be captured by simple Euclidean geometry, foundational to understanding the 720° topology requirement (Nölle, Suda, and Boxleitner, 2013).

⁵Nölle et al.’s work was presented at DSP 2013 (1–3 July 2013) as peer-reviewed conference proceedings, where they discuss Judd’s “super-importance of hue” and derive an effective hue-circle circumference of roughly 4π (720°) under their proposed metric.

$$\frac{s_2(S=1)}{s_1} = 4\sqrt{10} \approx 12.65 \approx 4\pi (= 720^\circ) \quad (2.3)$$

In a Euclidean space, the circumference of a unit circle of radius r is $2\pi r$. Here the derived ratio is approximately 4π , i.e., about $2.01 \times 2\pi$, motivating the informal statement that the hue circle behaves as if it had an effective “circumference” of roughly 720° rather than 360° (Nölle, Suda, and Boxleitner, 2013, p. 3).

2.3.2 Physical Interpretation

What does this mean physically? Consider walking around a hue circle at constant lightness and saturation, taking steps of one just-noticeable difference (JND). In a Euclidean space, you would return to your starting point after $2\pi r$ JND steps, where r is the radius. In the H2SI construction, the corresponding circumference-to-radius ratio is closer to 4π than 2π , i.e., approximately double in this sense.

This “extra length” cannot be represented without distortion in a single global three-dimensional Euclidean coordinate system. It suggests that perceptual color geometry is intrinsically curved, i.e., *non-Euclidean* (Nölle, Suda, and Boxleitner, 2013; Roberti and Peruzzi, 2023). Any attempt to embed color perception in a flat 3D coordinate system must introduce distortions, either uneven discrimination thresholds (as in CIELAB) or hue-dependent perceptual compression (as in HSV/HSL).

2.3.3 Implications for Existing Color Spaces

This result helps explain why even carefully designed spaces like CIELAB exhibit perceptual nonuniformities (Kong, 2021):⁶

- **CIELAB (1976):** Achieves approximate uniformity in spatial discrimination tasks but performs poorly as a predictor for temporal transitions. Kong (2021) notes explicitly: “CIELAB is not a useful space to predict the perception of dynamic colored light. Today, no color spaces are available that accurately predict the visibility of color differences over time” (Kong, 2021, p. 3).
- **CAM16-UCS:** Improved spatial uniformity through adaptive chromatic transforms, but computational cost limits real-time use

⁶Kong’s dissertation provides comprehensive evidence that CIELAB, while effective for spatial discrimination tasks, fails for temporal color perception: “CIELAB is not a useful space to predict the perception of dynamic colored light. Today, no color spaces are available that accurately predict the visibility of color differences over time” (Section 5.3, p. 530). Kong also confirms Sekulovski’s 10:1 asymmetry: “The threshold for L^* was found to be approximately 10 times smaller than for the chromaticity indices a^* and b^* .”

- **OKLab:** Optimized for practical uniformity within the constraint of Euclidean coordinates; excellent for gradients and interpolation
- **HSV/HSL:** Euclidean spaces with poor perceptual correspondence; hue angle does not represent equal perceptual steps

The engine adopts OKLab as a pragmatic compromise: a space that *minimizes* perceptual distortion while remaining computationally tractable (§2.4). The 720° result remains relevant for understanding why perfect uniformity is unattainable and why certain design choices (e.g., velocity weights, closure thresholds) require empirical calibration rather than pure geometric derivation.

2.3.4 Connection to Möbius Loop Topology

The 720° result has a surprising implication for loop closure strategies (§7.1–7.5). A Möbius strip, a surface with a single half-twist, also requires traversing 720° to return to the original orientation. This coincidence—both involving a 720° return—suggests a potentially useful design heuristic: Möbius-like operations (chromatic inversion with lightness continuity) may provide a compatible loop-closure strategy, though the precise mathematical relationship remains to be formalized.

Important caveat: This connection should be understood as a *design heuristic and analogy* rather than a formally derived theorem. The metric circumference result and perceptual nonuniformity are distinct phenomena from orientation reversal in a Möbius strip. The Möbius strategy (§7.4) exploits this parallel as a heuristic to achieve smooth loop closure.

Having established both the Riemannian curvature (§2.2) and the topological impossibility of perfect uniformity (§2.3), we now face a practical engineering question: which color space provides the best computational approximation? The answer lies in strategic compromise rather than theoretical perfection.

2.4 OKLab Color Space

2.4.1 Design Rationale

The selection of a working color space requires balancing competing objectives:

1. **Perceptual accuracy:** Euclidean distances should approximate perceptual differences as closely as possible

2. **Computational efficiency:** Transformations must be fast enough for real-time palette generation
3. **Numerical stability:** The space should behave well at gamut boundaries and extreme lightness values
4. **Industry validation:** The space should have independent verification and adoption

OKLab, introduced by Björn Ottosson in 2020 (Ottosson, 2020), achieves an excellent balance across these criteria. It has gained rapid adoption in web standards (CSS Color Level 4) (W3C CSS Working Group, 2022), creative software, and independent validation for gradient quality (Levien, 2021).

Table 2.1 synthesizes the trade-offs:

Table 2.1: Color space comparison for temporal color journey construction. *Uniformity* refers to how closely Euclidean distances match perceived differences (assessed via gradient smoothness and discrimination threshold alignment); *Cost* measures computational complexity; *Stability* indicates behavior at gamut boundaries and extreme lightness values. In practice, OKLab provides high-quality interpolation and gradients at low computational cost.

Color Space	Uniformity	Cost	Stability	Selected
sRGB	Poor	Low	Good	No
HSV/HSL	Poor	Low	Poor	No
CIELAB (1976)	Moderate	Moderate	Poor at extremes	No
CAM16-UCS	Excellent	High	Good	No
OKLab	Excellent	Low	Excellent	Yes

2.4.2 Coordinate Representation

A color in OKLab is represented by three coordinates (L, a, b) :

- L — **Perceived lightness** ranges from 0 (perceptual black) to 1 (brightest white). Equal differences in L correspond to approximately equal perceived brightness changes.
- a — **Green–red opponent axis** with positive values indicating reddish tints and negative values indicating greenish tints. Zero is achromatic along this axis.
- b — **Blue–yellow opponent axis** with positive values indicating yellowish tints and negative values indicating bluish tints. Zero is achromatic along this axis.

The axes are designed to be *approximately perceptually orthogonal*: altering L alone should not significantly change perceived hue or chroma; altering a or b primarily affects hue and saturation without changing perceived lightness (Ottosson, 2020).

The perceptual distance between two colors is computed as Euclidean distance:

$$\Delta E = \sqrt{(L_2 - L_1)^2 + (a_2 - a_1)^2 + (b_2 - b_1)^2} \quad (2.4)$$

where ΔE denotes color difference in OKLab perceptual space.⁷

2.4.3 Transformation Pipeline

The transformation from sRGB to OKLab follows a fixed matrix pipeline:

1. **Linearize sRGB:** Remove gamma encoding to obtain linear RGB values
2. **Linear sRGB to LMS-like:** Apply matrix M_1 to map linear sRGB into an LMS-like space (numerically fitted; not identical to physiological cone LMS)
3. **Nonlinear compression:** Apply a sign-preserving cube root to obtain $(l', m', s') = (\text{cbrt}(l), \text{cbrt}(m), \text{cbrt}(s))$
4. **LMS to Lab:** Apply matrix M_2 to obtain final (L, a, b)

The transformation matrices are (Ottosson, 2020)⁸:

$$M_1 = \begin{pmatrix} 0.4122214708 & 0.5363325363 & 0.0514459929 \\ 0.2119034982 & 0.6806995451 & 0.1073969566 \\ 0.0883024619 & 0.2817188376 & 0.6299787005 \end{pmatrix} \quad (2.5)$$

$$M_2 = \begin{pmatrix} 0.2104542553 & 0.7936177850 & -0.0040720468 \\ 1.9779984951 & -2.4285922050 & 0.4505937099 \\ 0.0259040371 & 0.7827717662 & -0.8086757660 \end{pmatrix} \quad (2.6)$$

These values are standardized in CSS Color Level 4 (W3C CSS Working Group, 2022) and match the reference implementation.

⁷All ΔE measurements in this specification refer to Euclidean distance in OKLab space unless explicitly noted otherwise. We use “color difference” and “perceptual distance” interchangeably when referring to this metric.

⁸The OKLab matrix coefficients and transformation pipeline are directly based on Ottosson’s C++ reference implementation, available under the MIT license at <https://bottosson.github.io/posts/oklab/>. The code was updated 2021-01-25 to use higher-precision sRGB matrices and exactly matching D65 values. For full details and code, see Appendix E.

Implementation note: Ottosson’s C++ code (MIT license) provides the canonical reference for these transformations. Depending on application, it may be preferable to use the sRGB matrices native to your environment for exact consistency. See Appendix E for the full code listing.

Each step models aspects of human vision: M_1 maps linear sRGB into an LMS-like response space fitted to color appearance data (so it is cone-inspired rather than a direct physiological LMS model); the cube root models nonlinear perceptual compression—a principle dating to Fechner’s seminal psychophysical work (as discussed in Fairchild 2013, Chapter 3) and refined through modern understanding of cone response functions (Ottosson, 2020); M_2 extracts lightness as a weighted average and opponent-color signals as cone differences.

2.4.4 Sign-Preserving Cube Root

Real implementations can encounter negative LMS values, particularly outside gamut or due to numerical precision at extreme parameter values. The engine uses the *standard sign-preserving cube root* as specified in the OKLab reference implementation: $\text{cbrt}(x) = \text{sgn}(x) \cdot |x|^{1/3}$. This ensures continuity across the achromatic axis and avoids internal inconsistencies when discussing gamut and unbounded behaviour.

2.4.5 Computational Efficiency

The OKLab conversion is computationally lightweight:

- **Operations:** 18 multiplications, 12 additions, 3 cube roots per conversion
- **Memory:** no heap allocations are required; the transform can be implemented as straight-line arithmetic plus a sign-preserving cube root
- **Parallelisation:** trivially vectorisable with SIMD instructions
- **Performance:** conversion overhead is negligible compared to palette generation logic

This efficiency supports real-time palette generation. The forward and inverse pipelines use fixed constant matrices with no iterative solving; any necessary gamut handling is addressed separately (§8.1).

2.4.6 Cartesian vs. Cylindrical Coordinates (OKLCh)

For certain operations, particularly hue manipulation, a cylindrical form is more convenient. OKLCh represents the same colors using:

L — **Lightness** identical to OKLab

C — **Chroma** defined as $C = \sqrt{a^2 + b^2}$, representing distance from the neutral axis (colorfulness/saturation)

h — **Hue angle** defined as $h = \text{atan2}(b, a)$, the angle around the a - b plane

The conversion is straightforward:

$$C = \sqrt{a^2 + b^2} \tag{2.7}$$

$$h = \text{atan2}(b, a) \tag{2.8}$$

The engine uses both representations internally:

- **Cartesian (OKLab)** for distance calculations, linear interpolation, and Bézier curve construction
- **Cylindrical (OKLCh)** for hue-related operations such as warmth bias (§5.1), complementary color calculation (§7.4), and palette mode operation (§6.2)

This separation allows clean reasoning about lightness dynamics (along L) versus chromatic dynamics (in the a - b plane or around h).

subsectionImplications for Journey Construction

The choice of OKLab has direct implications for color journey design:

1. **Linear interpolation produces smooth gradients:** A straight line between two colors in OKLab coordinates produces a perceptually smooth transition, unlike RGB, where perceptual step size varies substantially across the interpolation. Figure 2.1 illustrates this by comparing the perceptual distance between adjacent samples under linear RGB versus linear OKLab interpolation.
2. **Distance calculations are meaningful:** Constraints like $\Delta_{\min}$ and $\Delta_{\max}$ correspond to actual perceived differences, enabling principled constraint enforcement.
3. **Dynamics are separable:** The orthogonality of L , a , b means lightness can be adjusted independently of chromaticity, simplifying multi-objective optimization.
4. **Hue interpolation is stable:** Unlike HSV, where interpolating across the $0^\circ/360^\circ$ boundary causes discontinuities, OKLab’s Cartesian representation avoids such artifacts.

Because OKLab is unbounded, and can represent colors outside any physical display gamut, the engine must include explicit gamut mapping (§8.1).

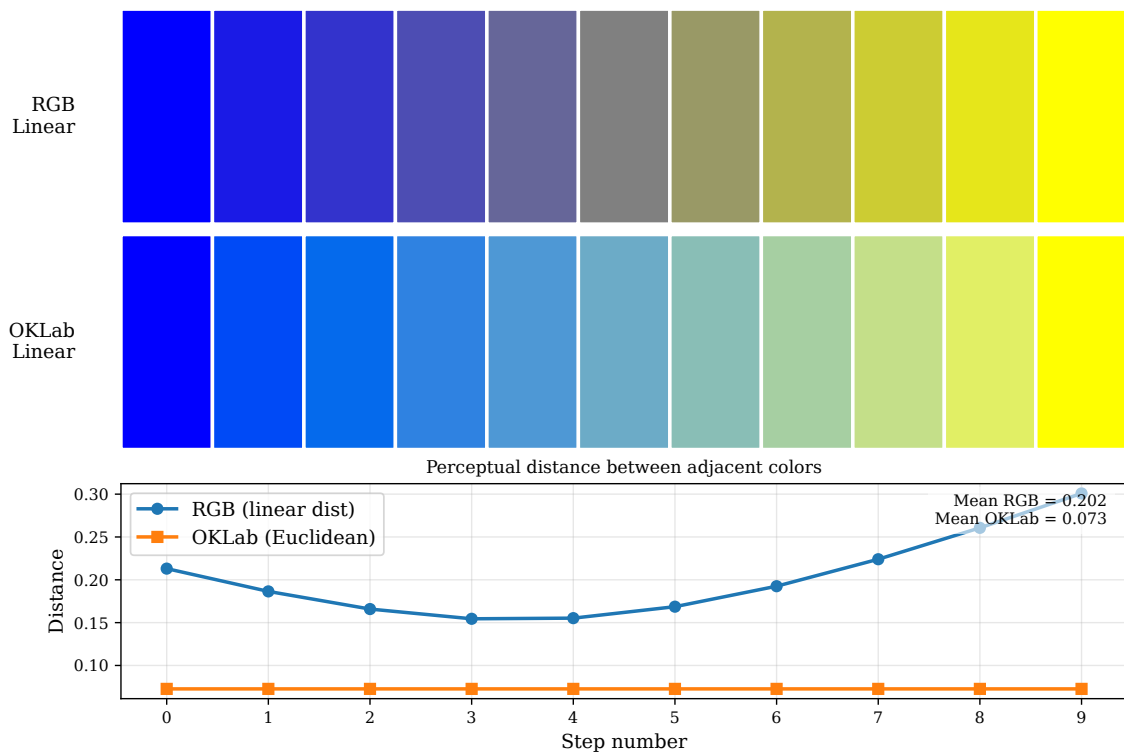


Figure 2.1: Perceptual distance between adjacent colors under linear interpolation. Linear RGB interpolation produces highly nonuniform perceptual steps, whereas linear interpolation in OKLab yields substantially more uniform step distances (lower mean and reduced variation), supporting OKLab as a practical approximation for perceptually smooth journeys.

While OKLab provides spatial perceptual uniformity, color journey design introduces a temporal dimension. As we will see, the human visual system processes color changes over time differently from spatial color differences, a distinction with important implications for velocity constraints.

2.5 Temporal vs. Spatial Color Perception

2.5.1 Eye Movement and Chromatic Sensitivity

Human color perception differs fundamentally between static spatial patterns and dynamic temporal transitions. Braun et al. (2017) measured chromatic sensitivity during different types of eye movements:

Smooth pursuit: When tracking a moving object with smooth eye movements, chromatic sensitivity *increases* by approximately 12% (Braun, Schütz, and Gegenfurtner, 2017).⁹ The visual system enhances color discrimination during continuous tracking.

Fixation: During stable fixation, chromatic sensitivity remains at baseline levels.

Saccadic suppression: During rapid eye movements (saccades), chromatic sensitivity *decreases* by approximately 58% (Braun, Schütz, and Gegenfurtner, 2017). The visual system suppresses color information during ballistic gaze shifts to avoid motion blur.

This asymmetry has direct implications for color journey design: transitions should be optimized for *smooth viewing* rather than rapid scanning. Gradual, continuous color changes exploit the enhanced chromatic sensitivity during pursuit, while abrupt changes risk perceptual suppression.

2.5.2 Temporal Asymmetry in Color Channels

Beyond eye movement effects, the visual system exhibits channel-specific temporal sensitivities. Sekulovski et al. (2007) conducted a series of psychophysical experiments to quantify these differences, using a classical method to measure discrete color difference thresholds.¹⁰

⁹These percentages are stated in the abstract of Braun, Schütz, and Gegenfurtner (2017).

¹⁰Sekulovski's research at Philips Research introduced the Delta-E-ab per second metric for temporal color transitions, establishing that smoothness thresholds are approximately 10 times smaller for luminance than for chroma/hue changes in CIELAB.

Experimental methodology

Sekulovski’s team presented observers with discrete linear temporal color transitions in CIE LCh space, varying independently along three axes: lightness (L), chroma (C), and hue (h). For each condition, they measured the threshold step size (expressed in ΔE_{ab} units) at which observers could still perceive the transition as perceptually smooth. They systematically varied:

- **Transition direction:** Three independent axes (lightness, chroma, hue)
- **Frequency:** Multiple frame rates (5, 10, 20, 30, and 40 Hz)
- **Base color:** Multiple starting colors with different chromaticity coordinates

Thresholds were analyzed using analysis of variance (ANOVA) with direction of change, frequency, and base color as fixed factors, and participant as a random factor (Sekulovski et al., 2007, pp. 113–114).

The 10:1 ratio: statistical findings

The ANOVA revealed a highly significant main effect of direction of change ($p < 0.01$) (Sekulovski et al., 2007, p. 113). Post hoc tests demonstrated that mean threshold values for lightness changes were approximately **10 times smaller** than those for chroma and hue changes (Sekulovski et al., 2007, p. 113). Quantitatively, this implies that to maintain comparable perceived smoothness, allowable step sizes in lightness should be about *one tenth* the size of step sizes in chroma or hue (in ΔE_{ab} units), reflecting a strong temporal asymmetry across channels.

Kong’s later dissertation (Kong, 2021, p. 3) summarizes the finding concisely:

“The visibility threshold of smoothness, i.e., the maximum color difference between two successive colors that is allowed in order to perceive a temporal color transition as smooth, is about ten times smaller for lightness changes than for chroma or hue changes in CIELAB” (Kong, 2021, p. 3).

Read operationally, the result indicates that lightness changes must be constrained much more tightly than chromatic changes for transitions to remain perceptually smooth.

In other words, observers are approximately **10 times more sensitive** to temporal changes in lightness than to changes in chromatic attributes. This 10:1 asymmetry is empirically robust, validated across multiple base colors and viewing frequencies by controlled experimental design and statistical analysis (Sekulovski et al., 2007, pp. 113–114).

Independence across base colors

Crucially, Sekulovski’s ANOVA also revealed that lightness thresholds showed *no significant interaction* with base color point, whereas chroma and hue thresholds did vary with base color (Sekulovski et al., 2007, p. 113). This finding is expressed as:

“No main effect of the base color point on the thresholds for lightness changes was found for the chromatic color points... This shows independence of the thresholds to changes in chromaticity and dependence on lightness” (Sekulovski et al., 2007, p. 113).

This independence across base colors is critical for color journey design: a single 10:1 temporal sensitivity weighting can be applied globally, rather than requiring color-dependent corrections. Within the tested conditions, the ratio appears stable across base colors, and we adopt it as a working default while retaining calibration as future work.

2.5.3 Design Implications

We model temporal smoothness as a weighted path-length constraint in OKLab/OKLCh, where weights approximate relative JND sensitivity per channel under transition viewing. The channel weights are chosen so that visually dominant changes (notably in lightness) are penalised more strongly than less salient changes. As Figure 2.2 summarizes, we implement this as a weighted path-length constraint in OKLab/OKLCh space.

These findings directly inform the engine’s velocity weighting scheme (§6.3). Figure 2.2 visualizes the intuition: small changes in lightness are typically more salient over time than comparable chromatic changes, motivating a substantially higher lightness weight.

1. **Lightness velocity weight:** $w_L = 10.0$ reflects the heightened temporal sensitivity (Sekulovski et al., 2007)
2. **Chroma velocity weight:** $w_C = 1.0$ serves as the baseline reference
3. **Hue velocity weight:** $w_H \approx 1.5$ to 2.0 is a design parameter requiring empirical validation

Justification for hue weight range

We treat w_H as a tunable parameter and adopt a conservative range (≈ 1.5 –2.0) rather than a single fixed value for three reasons. First, classic arguments about the “super-

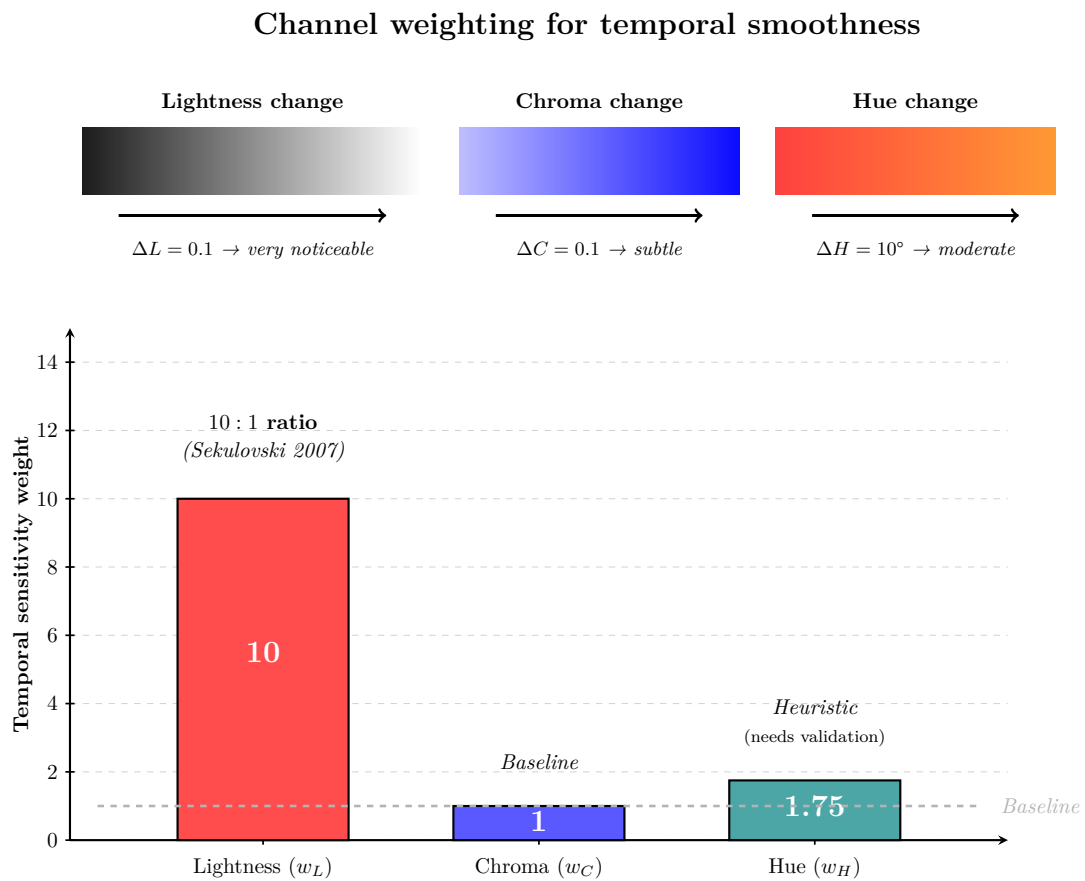


Figure 2.2: Temporal channel weighting for perceived smoothness. Lightness is up-weighted relative to chroma based on Sekulovski’s measured 10:1 temporal sensitivity asymmetry, while hue receives a moderate heuristic weight pending dedicated calibration (Sekulovski et al., 2007).

importance of hue” suggest that hue can contribute disproportionately to perceived difference around the chromatic circle relative to a purely Euclidean embedding (Judd, 1940; Nölle, Suda, and Boxleitner, 2013). Second, hue changes often correspond to a perceptually categorical change (e.g., “bluish” to “greenish”) even when chroma is held constant, so moderate weighting helps prevent visually abrupt phase shifts in temporal transitions. Third, existing temporal threshold studies support the strong $w_L : w_C$ asymmetry but do not provide an equally precise statistical separation between hue and chroma, so we avoid overclaiming and frame w_H as an empirically calibratable design choice.

The 10:1 ratio for $w_L : w_C$ is empirically validated through Sekulovski’s controlled psychophysical measurement and ANOVA analysis (Sekulovski et al., 2007, pp. 113–114). However, hue velocity weighting remains a *design heuristic*; Sekulovski’s data show hue and chroma thresholds are comparable in magnitude but do not provide a statistically precise ratio distinguishing them. Future work should employ psychometric methods (e.g., two-alternative forced-choice paradigms (Wang et al., 2022)) to measure hue temporal sensitivity rigorously and establish whether a hue-specific weighting factor can be justified beyond the baseline chroma sensitivity.

2.6 Summary: Perceptual Foundations Determine Engineering Constraints

The perceptual foundations established in this section impose four practical constraints on color journey construction:

1. **Work in perceptually uniform space:** The Riemannian structure (§2.2) requires a coordinate system where Euclidean interpolation approximates perceptual geodesics. OKLab provides this approximation with minimal computational overhead (§2.4).
2. **Accept imperfect uniformity:** The practical impossibility of perfect perceptual uniformity (§2.3) means all design parameters must be empirically calibrated rather than geometrically derived. The 720° result helps explain why certain operations, particularly loop closure, benefit from heuristic strategies.
3. **Weight lightness changes heavily:** The 10:1 temporal sensitivity ratio (§2.5) requires lightness velocity to dominate perceptual speed calculations. Lightness changes must be controlled more tightly than chromatic changes to maintain perceived smoothness.
4. **Optimize for smooth viewing:** Enhanced chromatic sensitivity during pursuit eye movements (§2.5) favors gradual, continuous transitions over abrupt changes. The engine prioritizes perceptual continuity over geometric optimality.

Where perceptual science remains unsettled (for example, suprathreshold distance additivity and hue-specific temporal weighting), we treat current choices as principled defaults and explicitly welcome future empirical updates.

2.6.1 Implications for Journey Construction

These perceptual constraints translate directly into algorithmic requirements:

Tunable parameters: Minimum step size $\Delta_{\min}$, maximum step size $\Delta_{\max}$, channel weights (w_L, w_C, w_H), and transition scaling factors all operate in OKLab perceptual units.

Calibration strategy: JND thresholds, temporal sensitivity ratios, and domain-specific viewing conditions determine default parameter values. User preferences can override defaults while maintaining perceptual validity.

Geometric machinery: The Riemannian structure justifies using Bézier curves, spline interpolation, and geodesic approximations in OKLab space (§3.5). Straight lines in OKLab approximate perceptually smooth paths.

Gamut management: Position-dependent metric tensors inform gamut mapping strategies: projection toward the achromatic center follows the intrinsic geometry revealed by discrimination threshold ellipses (§8.3).

With these perceptual foundations established, we now turn to the geometric and algorithmic machinery for constructing color paths that respect these constraints. Section 3.1 develops the computational framework for generating perceptually smooth color progressions within the perceptual space defined here.

Chapter 3

Journey Construction

3.1 The Journey Metaphor

The term “journey” was deliberately chosen over “gradient” or “palette” because it emphasises the *constructive process*—a path through color space—rather than the static result. The journey metaphor captures several key ideas:

Directionality There is a start and end (or a cycle)

Waypoints Anchors are places the journey must pass through

Path flexibility The route between waypoints can vary

Ordering Colors are sequenced with a sense of “before” and “after”

Critically, the journey is *not* a continuous function that callers sample arbitrarily. The engine outputs discrete swatches only. The continuous curve is an internal construction mechanism; callers receive N specific colors, not an interpolation function.

This distinction matters: callers should not expect to query “color at $t = 0.347$ ”. The engine can optimize internal representation freely, and behaviors like ping-pong playback become caller responsibility using the discrete output array.

3.2 Anchor Colors

Anchors are user-specified key colors that define the skeleton of the journey. They are guaranteed to appear in the output and serve as fixed boundary conditions for path construction.

Property	Specification
Minimum anchors	1
Maximum anchors	5
Input format	Hex, RGB, or OKLab
Internal representation	OKLab coordinates
Output guarantee	Each anchor appears exactly in output

The 5-anchor maximum is a deliberate design constraint informed by cognitive limitations and practical experience. Research on working memory suggests humans can reliably track 4 ± 1 items simultaneously (Cowan, 2001); beyond this, the perceptual “story” of a color journey fragments into disconnected segments. Additionally, cubic Bézier curves between consecutive anchors provide C^1 continuity guarantees (Farin, 2002), but as anchor count increases, maintaining smooth visual flow becomes increasingly difficult. This is a *design choice* balancing expressiveness against cognitive and computational complexity, not a technical limitation.

Anchors are processed in the order provided: $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_m$. The journey visits each anchor in sequence; for closed loops, it then returns to A_1 .

3.3 Mood Expansion (Single-Anchor Case)

When only one anchor is provided, the engine cannot construct a traditional transition between colors. Instead, it performs **mood expansion**—creating a set of harmonious colors centred on the single anchor.

To our knowledge, this single-anchor expansion approach represents a novel contribution. Traditional palette generators require at least two colors for interpolation; color harmony systems (complementary, triadic, analogous) generate geometrically-related hues but not perceptually-graded variations. Mood expansion bridges this gap by using the anchor’s inherent character to determine expansion direction.

Design Decision: Mood Expansion Direction

Choice: Single-anchor palettes expand along lightness-weighted directions in OK-Lab, using the anchor’s inherent character (hue, chroma, lightness) to determine expansion.

Rationale: Expanding in perceptually-coherent directions preserves the anchor’s “mood” or character. Grey anchors naturally expand along lightness (the only meaningful axis available); chromatic anchors expand toward complementary or analogous regions based on their position in color space.

Alternatives Considered:

- *Naive hue spin* — Rejected: produces arbitrary results disconnected from anchor character
- *Fixed expansion vectors* — Rejected: ignores anchor properties, loses mood coherence
- *Random sampling in OKLab* — Rejected: no perceptual relationship to anchor

Reference: §3.2, §3.5

Example: A dark navy anchor ($L = 0.25$, $h \approx 250^\circ$) expands primarily along the lightness axis toward lighter blues and teals, preserving the anchor’s cool character. A bright orange anchor ($L = 0.75$, $C = 0.15$) expands toward darker oranges and adjacent warm hues. The expansion direction emerges from the anchor’s inherent perceptual properties, not arbitrary geometric rules.

With one anchor A_1 , the engine:

1. Uses the anchor as the palette’s conceptual centre
2. Generates variations along multiple axes:
 - Lighter and darker variants (along L axis)
 - Slight hue rotations (around h in OKLCh)
 - Chroma variations (along C in OKLCh)
3. Forms a closed loop returning to the anchor

Single-anchor journeys are useful for generating monochromatic palettes with depth, creating UI state variations (hover, active, disabled) from a brand color, and animating “breathing” or “pulsing” color effects.

3.4 Multi-Anchor Paths

For m anchors ($m \geq 2$), the journey is composed of $m - 1$ segments:

$$\gamma_0 : A_1 \rightarrow A_2 \tag{3.1}$$

$$\gamma_1 : A_2 \rightarrow A_3 \tag{3.2}$$

$$\vdots \tag{3.3}$$

$$\gamma_{m-2} : A_{m-1} \rightarrow A_m \tag{3.4}$$

Each segment is constructed independently but with continuity constraints at junction points.

C^0 Continuity (Position). The end of segment γ_i equals the start of segment γ_{i+1} . This is automatically satisfied since both meet at anchor A_{i+1} .

C^1 Continuity (Tangent). For smooth transitions, the engine matches the direction of approach and departure at each anchor. This prevents “corners” in the color path.

By default, the engine applies smoothing at anchor junctions to achieve C^1 continuity, making the journey feel fluid rather than segmented. If sharp transitions are desired, this can be adjusted via the smoothness parameter (§5.3).

3.5 Bézier Curve Construction

Each segment is represented as a **cubic Bézier curve** in OKLab space Farin, 2002, providing flexibility, mathematical elegance, continuity control, and computational efficiency.

A cubic Bézier curve with endpoints P_0, P_3 and control points P_1, P_2 :

$$\gamma(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t) t^2 P_2 + t^3 P_3 \tag{3.5}$$

For $0 \leq t \leq 1$: $\gamma(0) = P_0$ (start anchor) and $\gamma(1) = P_3$ (end anchor).

The tangent vectors at endpoints are:

$$\gamma'(0) = 3(P_1 - P_0) \quad (\text{direction leaving start}) \tag{3.6}$$

$$\gamma'(1) = 3(P_3 - P_2) \quad (\text{direction arriving at end}) \tag{3.7}$$

Control Point Placement

By default, control points are placed on the line between anchors, making the Bézier equivalent to linear interpolation. Style parameters then “nudge” these control points to shape the curve:

- **Intensity** scales control point distance from the anchor line, adding curve drama
- **Temperature** biases the offset direction toward warm or cool hues
- **Smoothness** affects how aggressively tangents are matched at junctions

With default parameters, the engine produces simple linear interpolation—the most predictable baseline.

Ensuring C^1 Continuity

To match tangents at anchor A_{i+1} , the control point Q_1 of the next segment is placed such that:

$$Q_1 = A_{i+1} + (A_{i+1} - P_2) \quad (3.8)$$

This mirrors P_2 across the anchor, ensuring the outgoing direction matches the incoming direction.

3.6 Arc-Length Parameterisation

The Bézier parameter t does not correspond to distance along the curve (Piegl and Tiller, 1997; Kamermans, 2023). Equal increments in t produce unequal distances in OKLab space, leading to uneven perceptual steps—swatches cluster in curved regions where the curve bends sharply and spread apart in straighter sections.

This is a well-known property of parametric curves: the parameter t describes progress along the curve’s mathematical definition, not progress along its physical length. For perceptually uniform palette generation, we require *arc-length parameterisation*—sampling at equal distances along the actual path through color space.

The Arc-Length Solution

The engine estimates total arc length and samples at equal arc-length intervals (Levien, 2018):

1. **Estimate total arc length** — Numerically integrate $|\gamma'(t)|$ over $[0, 1]$
2. **Distribute sample points** — Place N samples at equal arc-length intervals
3. **Map back to t** — Find the t value corresponding to each arc-length position

Arc-length parameterisation ensures that output swatches are perceptually equidistant (in terms of path distance in OKLab), not just parametrically equidistant. Combined with OKLab’s perceptual uniformity, this produces swatches with consistent perceived differences between adjacent colors.

Practical Implementation

For efficiency, the engine uses numerical approximation following established techniques (Kamermans, 2023):

- Subdivide curve into many small segments (adaptive subdivision based on curvature)
- Sum Euclidean distances of segments to approximate total arc length
- Use binary search or Newton iteration to find t for target arc lengths
- Cache arc-length tables for repeated sampling of the same curve

The approximation error is bounded by the subdivision granularity. In practice, 100–200 subdivisions per segment provide sufficient accuracy for palette generation while maintaining computational efficiency.

Anchor Preservation

Anchors always appear exactly in the output sequence—they are not approximated or averaged. If the user provides anchors A, B, C and requests 7 swatches, the output includes A at position 0, C at position 6, and B at approximately the middle, with interpolated colors filling the gaps.

Chapter 4

Perceptual Constraints

4.1 Just-Noticeable Difference

The **Just Noticeable Difference (JND)** is the smallest color change that an average human observer can detect under standard viewing conditions. This concept originates from Weber’s and Fechner’s foundational psychophysics work and has been extensively studied in color science (Fairchild, [2013](#)).

In OKLab:

- Theoretical JND threshold: $\Delta E \approx 1.0$ unit (OKLab’s design target (Ottosson, [2020](#)))
- Practical threshold: $\Delta E \approx 2.0$ units (accounting for real-world variability)

The practical threshold of 2.0 units is derived from applying a safety margin to the theoretical JND. Color science literature consistently shows that laboratory JND values (measured under controlled illumination, neutral adaptation, and foveal viewing) underestimate the threshold needed for reliable discrimination in real applications (Fairchild, [2013](#); Luo, Cui, and Rigg, [2001](#)). Factors including ambient lighting variation, observer differences, display calibration, and peripheral viewing all increase the effective threshold. The $2\times$ multiplier is a conservative engineering choice, not an empirically-validated constant.

Notably, the CSS Color Level 4 gamut mapping algorithm uses a similar threshold concept: colors are considered “close enough” to clip when $\Delta E < 0.02$ on OKLab’s 0–1 scale (W3C CSS Working Group, [2022](#)), which corresponds to approximately 2 units on the 0–100 scale commonly used in color difference literature. Our threshold choice aligns with this industry practice.

ΔE Range	Perceptual Interpretation
< 1.0	Imperceptible to most observers
$1.0 - 2.0$	Barely perceptible, subtle
$2.0 - 3.0$	Noticeable, clear difference
$3.0 - 5.0$	Obvious difference, still smooth
> 5.0	Pronounced difference, may feel like a “step”

Note: This table represents design guidance derived from Fairchild Fairchild, 2013 and practical experience, not a universal standard. Thresholds vary significantly by application context.

4.2 $\Delta_{\min}$ Constraint (Distinguishability)

$$\Delta_{\min} \approx 2.0 \tag{4.1}$$

This is the **minimum allowed perceptual distance** between any two adjacent swatches.

Why 2.0? This value is set at the practical JND threshold (§4.1) to ensure that every color step is reliably perceptible across typical viewing conditions. Setting $\Delta_{\min}$ at 1.0 (the theoretical JND) would produce steps that are only *sometimes* distinguishable; the 2.0 threshold provides margin for the variability inherent in real-world color perception (Fairchild, 2013). There is no value in generating colors that viewers cannot reliably distinguish—it wastes palette capacity and creates the impression of redundancy.

Enforcement

If two anchors are closer than $\Delta_{\min}$:

- **Collapse case:** The engine may treat them as essentially the same anchor, producing no intermediate swatches
- **Skip intermediates:** No interpolation is performed; the journey jumps directly from one to the other

When anchors are extremely close ($\Delta < \Delta_{\min}$), the engine does not attempt to create meaningless intermediate colors.

4.3 $\Delta_{\max}$ Constraint (Coherence)

$$\Delta_{\max} \approx 5.0 \tag{4.2}$$

This is the **maximum allowed perceptual distance** between any two adjacent swatches.

Why 5.0? This value represents the upper bound of “comfortable” color transitions—steps large enough to be clearly distinct but not so large as to feel discontinuous. The 5.0 threshold emerges from the JND interpretation table (§4.1): differences above ~ 5 units cross from “obvious but smooth” into “pronounced step” territory. This is a design heuristic informed by practical experimentation with generated palettes, not a value derived from formal perceptual studies. Alternative implementations may choose different thresholds based on their aesthetic goals.

Enforcement

If a segment between anchors has total distance $D > \Delta_{\max}$:

1. **Subdivide:** The segment is divided into n sub-segments where $n = \lceil D/\Delta_{\max} \rceil$
2. **Insert intermediates:** $n - 1$ intermediate swatches are inserted
3. **Cap subdivisions:** $n \leq 5$ to prevent excessive palette length

Why cap at 5? This limit prevents runaway palette growth when anchors are extremely far apart in color space. The value 5 is chosen to align with the anchor count limit (§3.2)—maintaining cognitive tractability. With at most 5 intermediates between any two anchors, users can reason about the palette structure. If anchors are extremely far apart ($D > 25$ units), each step will exceed $\Delta_{\max}$, which is acceptable: the user has explicitly requested a large color jump by their anchor selection.

4.4 Adaptive Sampling

Given anchors A and B with perceptual distance D , the adaptive sampling algorithm:

1. If $D < \Delta_{\min}$: append only B to output (skip intermediates)
2. Otherwise: compute $n = \min(5, \lceil D/\Delta_{\max} \rceil)$
3. For $i = 1$ to n : interpolate at $t = i/n$ and append to output

Each segment has length D/n , guaranteeing:

- Each step $\leq \Delta_{\max}$
- Each step $\geq \Delta_{\min}$ (approximately, given the constraints)

Total Distance D	Segments n	Step Size D/n
3.0	1	3.0
7.0	2	3.5
12.0	3	4.0
20.0	4	5.0
30.0	5 (capped)	6.0

When D is very large and capped at $n = 5$, step sizes may exceed $\Delta_{\max}$. This is a trade-off for palette manageability.

4.5 Constraint Enforcement

Some anchor configurations make constraints impossible to satisfy fully—two anchors very close together cannot yield many distinguishable steps; requesting 100 swatches between nearby colors is unsatisfiable.

The engine prioritises producing output over strict enforcement:

1. **Best effort:** The engine produces the closest valid palette
2. **Diagnostics:** Constraint violations are reported (not thrown as errors)
3. $\Delta_{\min}$ **priority:** Distinguishability takes precedence—colors below JND are never output

Distance constraints are evaluated on the actual path (after dynamics and gamut mapping are applied), not the straight-line distance between anchors. If dynamics increase path length, more intermediate swatches may be needed; if gamut clipping shortens effective distances, fewer may suffice.

The Perceptual Comfort Zone

The ideal step size falls in the range $2.0 \leq \Delta E \leq 5.0$ —steps are clearly distinguishable but not jarring. This is the “sweet spot” where palettes feel both coherent and varied.

Chapter 5

User-Facing Style Controls

5.1 Temperature Control

Temperature biases the journey’s hue path toward warm or cool colors (Hunt, 2004). This control operates in the hue dimension of OKLCh (§2.4.6), biasing the path selection when multiple hue routes are possible.

Range	−1.0 (cool) to +1.0 (warm), default 0.0
Effect	Shifts interpolation path through warm or cool hue regions

When interpolating hue in OKLCh, there are always two possible paths around the color wheel. By default, the engine takes the shortest path. The temperature parameter overrides this:

- **Positive values** (+0.1 to +1.0): Bias toward warm hues (reds, oranges, yellows)
- **Negative values** (−0.1 to −1.0): Bias toward cool hues (blues, cyans, greens)
- **Zero**: Take the shortest hue path (default)

The same anchor pair can produce vastly different palettes: purple-to-green with temperature 0 passes through blue (cool, oceanic), while temperature +1 passes through red and yellow (warm, sunset-like).

Temperature is implemented via direction blending:

$$\vec{v} = (1 - \alpha|T|) \cdot \hat{v}_{\text{base}} + \alpha|T| \cdot \hat{v}_{\text{warm/cool}} \quad (5.1)$$

where $\hat{v}_{\text{base}}$ is the shortest-path direction, $\hat{v}_{\text{warm/cool}}$ is the target hue direction, and $\alpha \in (0, 1)$ is a blending coefficient (typically $\alpha = 0.8$) that ensures partial preservation of the base direction, preventing jarring snaps to pure warm or cool paths.

5.2 Intensity Control

Intensity controls the chromatic drama of the journey—how far control points deviate from straight-line interpolation.

Range	0.0 (minimal) to 1.0 (maximum), default 0.5
Effect	Scales Bézier control point offset from the anchor line

- **Low intensity** (0.0–0.3): Subtle, nearly linear paths; control points close to the anchor line
- **Medium intensity** (0.4–0.6): Moderate curves with balanced chromatic excursions
- **High intensity** (0.7–1.0): Dramatic arcs; pronounced curves through color space

Mathematically, intensity scales the control point offset:

$$P_1 = \text{lerp}(P_0, P_3, \tfrac{1}{3}) + \iota \cdot \vec{s}_1 \quad (5.2)$$

$$P_2 = \text{lerp}(P_0, P_3, \tfrac{2}{3}) + \iota \cdot \vec{s}_2 \quad (5.3)$$

where ι is the intensity parameter and $\vec{s}_1, \vec{s}_2$ are style-determined offset vectors (typically into higher-chroma regions).

Intensity affects path geometry, not endpoint colors. The anchors remain unchanged (§3.2); only the route between them becomes more or less dramatic. This is achieved by scaling the Bézier control point offset (§3.5).

5.3 Smoothness Control

Smoothness controls the strength of C^1 continuity enforcement at anchor junctions.

Range	0.0 (sharp) to 1.0 (smooth), default 0.7
Effect	Controls transition character at anchor points

- **High smoothness** (0.8–1.0): Full tangent matching; the journey flows smoothly through anchors
- **Medium smoothness** (0.4–0.7): Partial tangent matching; gentle rounding at junctions
- **Low smoothness** (0.0–0.3): Minimal smoothing; sharper transitions at anchors

High smoothness is appropriate for gradients and animations where flow matters. Low smoothness may be preferred when anchors represent distinct states and clear transitions are desired.

Smoothness also affects perceived velocity consistency—higher values produce more uniform perceptual change rates along the journey.

5.4 Secondary Parameters

Beyond the primary triad (temperature, intensity, smoothness), several secondary parameters provide finer control:

Lightness Bias. Range -1.0 to $+1.0$, default 0 . Shifts all colors lighter (positive) or darker (negative). Applied proportionally to avoid clipping:

$$L' = \begin{cases} L + \lambda(1 - L) & \text{if } \lambda > 0 \\ L + \lambda \cdot L & \text{if } \lambda < 0 \end{cases} \quad (5.4)$$

Chroma. Range 0.0 to 2.0 , default 1.0 . Multiplies the chroma (saturation) of all colors. Values > 1 increase saturation; values < 1 desaturate toward greyscale.

Contrast. Range 0.0 to 2.0 , default 1.0 . Expands or compresses the lightness range. High contrast darkens darks and lightens lights; low contrast pushes toward mid-grey. Implemented via S-curve:

$$f_L(\alpha) = \frac{\alpha^p}{\alpha^p + (1 - \alpha)^p} \quad (5.5)$$

where p is derived from the contrast parameter.

Vibrancy. Range 0.0 to 2.0 , default 1.0 . Selectively boosts chroma of less-saturated colors while leaving already-vibrant colors unchanged (Poynton, 2012). Useful for counteracting desaturation in complementary-color interpolation.

$$C' = C \cdot \left(1 + v \cdot \left(1 - \frac{C}{C_{\max}} \right) \right) \quad (5.6)$$

where $C_{\max}$ is the maximum achievable chroma at the color's current lightness L and hue h within the sRGB gamut boundary (see §8.1 for gamut constraints).

5.5 Parameter Interactions

Dynamics parameters are designed to be *largely independent*:

- Lightness affects only L
- Chroma affects only C
- Temperature affects only hue path
- Contrast affects L distribution, not absolute values

However, some combinations can conflict:

Combination	Potential Issue
High chroma + low lightness	May exceed gamut
High contrast + extreme anchors	May clip at black/white
High temperature + cool anchors	Long hue path, more colors needed

The engine applies all dynamics as specified, then performs gamut mapping (§8.1–§8.4). Conflicts are resolved by clamping to valid values, not by preventing the combination. Diagnostics report when settings caused gamut corrections (§10.4).

With all parameters at default values, the engine produces simple linear interpolation—the most predictable baseline. Presets encode tested combinations that work well together for specific aesthetic goals.

Chapter 6

Modes of Operation

6.1 Journey Mode

Journey Mode is the default mode of operation, optimising for **smooth, ordered sequences** suitable for timelines, gradients, transitions, and contexts where perceptual continuity matters. It leverages the full journey construction pipeline (§3.1–§3.6) with perceptual constraints (§4.1–§4.5).

Aspect	Behavior
Adjacency constraint	Enforces $\Delta_{\min} \leq \Delta(C_i, C_{i+1}) \leq \Delta_{\max}$
Path construction	Smooth Bézier curves through anchors
Continuity	C^1 continuity at anchor junctions
Sampling	Arc-length parameterised for perceptual uniformity

Journey Mode is appropriate for:

- Timeline visualisations (progress indicators, history)
- State transitions (hover, active, disabled)
- Mood-based palettes (warm sunset, cool ocean)
- Animation color cycling
- Any context where “flow” matters

Journey Mode is *not* appropriate for categorical data where each color represents a distinct entity, maximum contrast requirements, or semantic color mapping (red=danger, green=safe).

6.2 Categorical Mode

Categorical Mode optimizes for **maximum distinguishability** between swatches, suitable for discrete data categories, legends, and accessibility-critical applications (Fairchild, 2013). Unlike Journey Mode, which optimizes adjacent-color relationships (§4.2), Categorical Mode optimizes all pairwise relationships.

Aspect	Behavior
Distance goal	Maximise perceptual distance between all pairs
Path construction	May use hue maximisation, not smooth curves
Continuity	Not prioritised
Output order	Less meaningful than in Journey Mode

Instead of constraining adjacent distances, Categorical Mode maximises minimum pairwise distance:

$$\max \min_{i \neq j} \|C_i - C_j\|_{\text{OKLab}} \quad (6.1)$$

subject to staying within displayable gamut and respecting any semantic constraints.

Aspect	Journey Mode	Categorical Mode
Curve type	Smooth Bézier	May use angular paths
Sampling	Sequential along path	Optimized spacing in color space
Anchor role	Waypoints on path	Fixed category colors
Output order	Meaningful sequence	Order less important

Categorical Mode is appropriate for pie charts, bar charts with distinct categories, legend colors, and status indicators (error, warning, success, info).

6.3 Perceptual Velocity

Beyond perceptual *distance* (how different two colors are), **perceptual velocity** addresses how *fast* color change feels along a sequence.

Fast hue swings feel “faster” or more noticeable than the same ΔE spent on subtle lightness shifts. The *type* of change affects perceived speed, not just the magnitude. This observation is consistent with color appearance research showing that hue changes are more salient than equivalent lightness or chroma changes (Fairchild, 2013).

Perceptual velocity v can be modelled as a weighted sum:

$$v = w_L \cdot \frac{dL}{dt} + w_C \cdot \frac{dC}{dt} + w_h \cdot \frac{dh}{dt} \quad (6.2)$$

The following weights are *design heuristics* derived from practical experimentation, not empirically-validated constants:

Dimension	Relative Weight	Perceptual Effect
Lightness (L)	1.0 (baseline)	Subtle, gradual feel
Chroma (C)	~ 1.2	Moderate drama
Hue (h)	$\sim 1.5\text{--}2.0$	High impact, “fast” feel

Note: These weights reflect observed behavior in prototype testing. Formal psychophysical validation remains future work.

Velocity is relevant for temporal applications:

- **Calm presets** prefer L/C changes over hue (low velocity)
- **Energetic presets** use more hue swings (high velocity)
- **Rhythmic effects** alternate high/low velocity regions

Influencing Velocity

Users do not directly set velocity weights—they are internal heuristics. Instead, perceptual velocity emerges from the interaction of style parameters with anchor placement:

- **High temperature** increases hue travel between anchors, raising perceived velocity
- **High intensity** creates more curved paths with greater chromatic excursion
- **Anchors with similar hues** produce low-velocity journeys dominated by lightness changes
- **Anchors spanning the color wheel** produce high-velocity journeys with rapid hue shifts

Presets encode velocity profiles implicitly: “calm” presets (**smooth**, **muted**) prefer parameters that minimise hue change, while “energetic” presets (**vivid**, **neon**) encourage chromatic drama.

6.4 Mode Selection Guidelines

Use Case	Recommended Mode
Output for discrete categories?	Categorical
Need smooth animation?	Journey + closed/pingpong loop
Need timeline colors?	Journey + open path
Need mood palette from single color?	Journey + single anchor
Need maximum contrast?	Categorical with high separation

Hybrid Approaches

Some applications benefit from combining modes:

1. **Categorical anchors, journey fill:** Use Categorical Mode to select maximally distinct anchors, then Journey Mode for smooth transitions between them
2. **Journey with velocity control:** Use Journey Mode but adjust sampling based on velocity for rhythmic effects
3. **Mode switching:** Different parts of an application use different modes (categorical for legend, journey for timeline)

Chapter 7

Loop Strategies

The previous sections established *what* the engine optimizes for—smooth sequential flow in Journey Mode or maximum categorical distinction—and *how* style controls shape the journey’s character. This chapter addresses a complementary concern: *what happens at the boundaries* when a palette must extend beyond its natural start-to-end path, whether through cycling, reversal, or progressive evolution.

Five loop strategies provide different behaviors for sequences that need to repeat or continue beyond a single traversal.

7.1 Open Strategy

The **open** strategy is the default: the journey proceeds from the first anchor to the last anchor and stops. There is no attempt to connect the end back to the beginning.

$$J(t) : t \in [0, 1] \quad \text{where } J(0) = A_1, \quad J(1) = A_m \quad (7.1)$$

The output is a finite sequence with no wrap consideration. If looped externally by the caller, there will be a visible jump from A_m back to A_1 .

Open paths are appropriate for one-time state transitions, linear progress indicators, and palettes not intended to cycle.

7.2 Closed Strategy

The **closed** strategy forms a complete cycle, returning smoothly from the last anchor back to the first (Farin, [2002](#)):

$$J(0) = J(1) = A_1 \quad (7.2)$$

The engine adds a segment from A_m back to A_1 , creating a closed path with C^1 continuity at the wrap point. The tangent leaving A_m toward A_1 matches the tangent arriving at A_1 from A_m , preventing a “corner” in color space.

In closed loop mode, the output array *omits the duplicate final color*. The last swatch is adjacent to (but not identical to) A_1 , because A_1 is already the first swatch. When looped, the sequence wraps seamlessly.

Closed loops are appropriate for continuous color cycling animations, hue wheels, and looping ambient effects.

7.3 Ping-Pong Strategy

The **ping-pong** strategy proceeds from start to end, then reverses back to start, creating back-and-forth oscillation.

For parameter $u \in [0, 2]$:

$$\tilde{t} = \begin{cases} u & 0 \leq u < 1 \\ 2 - u & 1 \leq u < 2 \end{cases} \quad (7.3)$$

No new colors are generated in the reverse pass—the path is exactly retraced. Smooth reversal occurs at the turning point (A_m) and smooth loop at the return point (A_1).

Ping-pong output includes the forward sequence; callers can traverse the array forward then backward, or the engine can output the full forward-and-back sequence explicitly.

Ping-pong is appropriate for “breathing” or “pulsing” color effects, bidirectional progress indicators, and oscillating state visualisations.

7.4 Möbius Strategy

The **Möbius** strategy is a half-twist loop requiring *two complete traversals* to return to the starting color—named after the Möbius strip, a surface with a half-twist where walking one complete loop brings you to the “other side,” requiring a second loop to return to your starting position.

In color terms: after one cycle through the journey, you arrive at a color *related to* but distinct from the start; only after the second cycle do you return to the original. This creates animations that feel subtly different on alternating cycles.

One approach inverts the chromatic components at the halfway point:

$$J_{\text{mobius}}(1) = (L, -a, -b) \quad \text{when } J_{\text{mobius}}(0) = (L, a, b) \quad (7.4)$$

This yields the complementary color (180° hue rotation) at the midpoint. The viewer perceives continuous change that takes two “laps” to truly repeat:

- Cycle 1: Journey from Red $\rightarrow \dots \rightarrow$ Cyan (complement)
- Cycle 2: Journey from Cyan $\rightarrow \dots \rightarrow$ Red (back to original)

Möbius is an advanced option requiring careful understanding of the twist behavior. It is appropriate for complex animations with implicit alternation, effects that should vary between loops, and artistic/generative applications.

7.5 Phased Strategy

The **phased** strategy applies a systematic shift each repetition, so the palette evolves over time rather than repeating exactly.

On cycle k :

$$J_k(t) = J_0(t) + k \cdot \text{shift} \quad (7.5)$$

where the shift is a per-cycle offset vector in OKLab space. Typical configurations specify shifts as hue rotations in degrees (e.g., +10° per cycle, applied in OKLCh) or lightness adjustments in OKLab L units (e.g., +0.05 per cycle). The shift magnitude and dimension are configuration parameters.

Visual effect:

- Cycle 1: Blue $\rightarrow$ Green
- Cycle 2: Blue+10° $\rightarrow$ Green+10° (slightly shifted hues)
- Cycle 3: Blue+20° $\rightarrow$ Green+20°
- ...eventually wraps around

Phased loops are parameterised by shift amount and dimension. This is the most complex loop type, appropriate only when continuous evolution is specifically desired—slowly evolving animations, generative art with progressive change, and long-running visualisations that should not feel static.

7.6 Output Semantics

Design Decision: Unrolled Loop Output

Choice: All loop strategies output a flat array of N colors. Loops are “unrolled” into a sequential list with no nested structure or cycle metadata.

Rationale: Flat arrays are universally consumable across languages and use cases. Callers needing cycle information can compute it from strategy and count; most callers don’t need it.

Alternatives Considered:

- *Segmented output (array of cycles)* — Rejected: adds complexity for minority use case
- *Cycle metadata in output* — Rejected: bloats output, caller can derive if needed
- *Iterator/generator model* — Rejected: not portable across language bindings

Reference: §10.3, Appendix D

Non-Repetition Rule

Loop outputs omit duplicate colors at boundaries. If the loop returns to the starting color, that color is *not* repeated at the end of the array:

- Conceptual path: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A$
- Output array: $[A, B, C, D, E]$ (5 elements)
- When looped: $A, B, C, D, E, A, B, C, D, E, \dots$ (A follows E naturally)

This prevents doubled colors when looping externally and reduces array length.

Caller Responsibilities

Simple repetition behaviors are caller responsibility, not engine features. The engine provides smooth loop paths and appropriate swatch sequences; the caller controls “play N times then stop”, animation timing, and pause/resume.

Chapter 8

Gamut Management

With path construction, perceptual constraints, style controls, and loop strategies established, one critical concern remains: ensuring that every color the engine generates is actually displayable. OKLab’s perceptual uniformity comes at a cost—it can represent colors that no physical display can reproduce. This chapter addresses gamut management: the techniques for keeping generated colors within the bounds of real displays.

8.1 The Gamut Boundary Problem

OKLab is a perceptually uniform color space, but it encompasses colors that lie outside any real display’s capabilities (Morovič, 2008):

- **High-chroma colors** at certain hue angles exceed sRGB
- **Very dark or very light saturated colors** clip
- **Certain hue regions** (cyan-blue, yellow-green) are particularly constrained

When constructing journeys, naive Bézier curves may traverse impossible colors. Hard clipping (clamping RGB values) creates visible discontinuities and hue shifts. Perceptual uniformity breaks down at gamut boundaries.

The fundamental constraint: all journey colors must satisfy

$$\gamma(t) \in G_{\text{sRGB}} \subset \text{OKLab} \quad \forall t \quad (8.1)$$

where G_{sRGB} represents the sRGB gamut volume mapped into OKLab space.

8.2 Design-Time Gamut Awareness

The first layer of gamut management is **prevention**—constructing paths that stay within gamut.

When placing Bézier control points, the engine:

1. **Prefers moderate chroma levels** — Control points default to $\sim 70\text{--}80\%$ of maximum chroma at their hue/lightness
2. **Detects candidate out-of-gamut points** — Before finalising control points, checks sRGB validity
3. **Pulls toward lower chroma** — If out-of-gamut detected, reduces chroma while preserving hue and lightness
4. **Uses gamut envelope models** — Employs OKLCh gamut boundary approximations for efficient checking

This proactive approach reduces the likelihood of out-of-gamut samples, preserving design intent where possible.

8.3 Gamut Correction

The second layer handles residual out-of-gamut colors through **soft correction** at sampling time. Our approach aligns with the CSS Color Level 4 gamut mapping algorithm (W3C CSS Working Group, 2022), which has been implemented in reference libraries such as Color.js (Verou and Lilley, 2024).

For a candidate color in OKLCh (L, C, h) :

1. Check sRGB validity via conversion
2. If out-of-gamut, perform binary search for maximum valid chroma at that (L, h)
3. Return the corrected color with reduced C , preserved L and h

The binary search converges rapidly (typically 10–15 iterations) because chroma is monotonically related to gamut membership at fixed lightness and hue. The CSS standard specifies terminating the search when $\Delta E < 0.02$ (on OKLab’s 0–1 scale), indicating the corrected color is perceptually indistinguishable from the gamut boundary (W3C CSS Working Group, 2022).

Design Decision: Two-Layer Gamut Handling

Choice: Gamut management uses two layers: (1) design-time prevention via gamut-aware control point placement, and (2) sample-time correction via chroma reduction with hue preservation.

Rationale: Post-correction alone can introduce perceptual discontinuities when many samples need large corrections. Prevention minimises corrections needed; correction handles residual cases gracefully. This approach follows established best practices in color management (Morovič, 2008).

Alternatives Considered:

- *Hard clipping (RGB clamping)* — Rejected: destroys hue, creates visual artifacts
- *HSV/HSL fallback* — Rejected: breaks perceptual uniformity guarantee
- *Post-processing only* — Rejected: large corrections cause discontinuities
- *Hue rotation to fit gamut* — Rejected: changes color identity unacceptably

Reference: §8.2, §8.4

8.4 Hue Preservation

When gamut mapping is required, the engine follows a strict priority hierarchy rooted in established gamut mapping practice (Morovič, 2008):

1. **Hue is sacred** — Never shift hue to fit gamut
2. **Lightness is strongly preserved** — Only adjust if absolutely necessary
3. **Chroma absorbs the compromise** — Reduce saturation to fit

This hierarchy is not arbitrary: research on gamut mapping algorithms shows that hue shifts are perceptually more objectionable than chroma reductions (Morovič, 2008; Fairchild, 2013). Humans are particularly sensitive to hue changes in familiar objects (skin tones, sky, foliage), and even small hue shifts can appear “wrong” while equivalent chroma reductions appear merely “less vivid.”

This preserves the journey’s “character” (its hue story) even when display limitations require desaturation (Hunt, 2004). A “sunset palette” remains warm-hued even if some oranges must be less saturated.

Interaction with Style Controls

Dynamic controls affect gamut behavior:

Control	Gamut Implication
Chroma > 1.0	Increases gamut pressure; more mapping likely
Chroma < 1.0	Reduces gamut pressure; safer
Vibrancy high	Pushes mid-journey toward gamut edges
Lightness extreme	May limit available chroma range
Temperature shift	Changes which hue regions are under pressure

If dynamics push colors out of gamut, chroma is the first to be moderated. Diagnostics report when dynamic settings caused gamut corrections.

Diagnostics

The engine reports gamut-related information:

- Number of colors requiring correction
- Maximum chroma reduction applied
- Which anchors were at gamut boundaries

This allows callers to understand why output differs from the theoretical ideal and adjust anchors or dynamics to avoid corrections.

Chapter 9

Variation and Determinism

The preceding chapters define the engine’s functional behavior: how it constructs paths, applies constraints, responds to style controls, handles loops, and manages gamut boundaries. This chapter addresses a cross-cutting concern that underlies all of these capabilities: the requirement for deterministic, reproducible output—and the controlled variation layer that enables exploration without sacrificing predictability.

9.1 Determinism Requirement

The Color Journey Engine *must* be programmatically deterministic. Given the same inputs, it *must* produce the same outputs, up to small numerical differences ($\sim 0.2\%$) arising from floating-point arithmetic and platform differences. This requirement enables the “config as ID” pattern central to the API philosophy (§10.1).

This is a hard requirement, not a preference:

- **Debugging** — Developers must be able to reproduce bugs
- **Testing** — Automated tests need predictable outputs
- **Sharing** — Configurations can be shared as compact seeds/parameters
- **Version control** — Generated assets can be diffed meaningfully
- **Caching** — Identical inputs can be cached without recomputation

For identical output, the following must match: anchor colors (same order and representation), all configuration parameters, variation mode and seed (if used), and implementation version.

No source of non-determinism may affect palette generation: system time, thread scheduling, external state, true random numbers, hash table iteration order, and uninitialised memory are all prohibited.

9.2 Controlled Variation

While determinism is required, users may want to generate alternatives, add organic texture, or explore nearby palettes. The **variation layer** provides this through controlled, seeded perturbations.

Mode	Effect
<code>off</code>	No variation; pure interpolation
<code>subtle</code>	Small perturbations; $\Delta E < 1$
<code>noticeable</code>	Moderate perturbations; ΔE 1–2
<code>extreme</code>	Large perturbations; ΔE 2–4

Variation modes are categorical rather than continuous, preventing the paralysis of infinite choice and providing meaningful distinct options.

Variation adds small offsets to interpolated colors in OKLab space:

$$L' = L + \delta_L \tag{9.1}$$

$$a' = a + \delta_a \tag{9.2}$$

$$b' = b + \delta_b \tag{9.3}$$

where perturbations are drawn from a Gaussian distribution with standard deviation scaled by mode.

Anchors are *never* perturbed—variation applies only to interpolated colors. This ensures user-specified key colors remain exact (§3.2). Perturbations taper near anchors (scaling by $\sin(\pi t)$) for smooth transitions.

9.3 Seed Handling

The variation layer uses a **pseudo-random number generator (PRNG)** with these properties (Knuth, 1997):

- **Seeded** — Initialised from user-provided seed
- **Deterministic** — Same seed produces same sequence
- **Well-distributed** — Numbers uniformly distributed
- **Specified algorithm** — Implementation uses documented algorithm (e.g., xoshiro256** (Blackman and Vigna, 2021))

Seed Value	Behavior
Explicit integer	Use that seed
<code>null</code> or omitted	Default seed (0) or variation disabled

If variation is requested but no seed provided, the engine uses a *default seed (0)*, not a random seed. This preserves determinism. If you want randomness, you must generate and provide the seed.

The order in which random numbers are consumed (per-color L, a, b perturbations) is fixed and documented—changing advancement order would change outputs for existing seeds.

9.4 Reproducibility

With determinism, a configuration object serves as a unique identifier for a specific palette. This enables the “config as ID” pattern:

- Store the configuration instead of the full palette
- Transmit compactly between systems
- Regenerate on any conforming implementation

Users can explore alternatives by iterating seeds: `seed=1` produces variant A, `seed=2` produces variant B, and so on. Each seed deterministically produces a unique but related palette.

Full reproducibility is guaranteed within the same engine version. Version changes may affect output (documented as breaking changes in semantic versioning). Recommended practice: store full configuration with generated palettes for future reference.

Chapter 10

API Design and Output Structure

10.1 API Philosophy

The engine is a **pure function** that transforms configuration into palette (Bloch, [2008](#)):

$$(\text{anchors}, \text{config}) \rightarrow \text{palette}$$

Key characteristics:

- **Stateless** — No memory between calls
- **Deterministic** — Same inputs produce same outputs
- **Focused** — Does one thing well (color journey generation)
- **No side effects** — No global state, no external dependencies

The guiding principle: the engine should do only what the caller cannot do themselves. Complexity belongs in construction, not in the public interface (Gamma et al., [1994](#)).

10.2 Input Parameters

Parameter	Type	Required	Description
<code>count</code>	integer	Yes	Number of colors to generate
<code>anchors</code>	array	Yes	1–5 colors (hex, RGB, or OKLab)
<code>temperature</code>	float	No	–1 to +1, default 0
<code>intensity</code>	float	No	0 to 1, default 0.5
<code>smoothness</code>	float	No	0 to 1, default 0.7
<code>mode</code>	enum	No	<code>journey</code> or <code>categorical</code>
<code>loop</code>	enum	No	<code>open</code> , <code>closed</code> , <code>pingpong</code> , etc.
<code>seed</code>	integer	No	Variation seed (omit for no variation)
<code>preset</code>	string	No	Named preset (expands to parameters)

When a preset is specified, it expands to a parameter set. Explicit parameters override preset defaults.

10.3 Output Structure

The engine returns discrete swatches (not continuous functions) with three components:

Palette Array. An ordered list of color swatches, each containing:

- `hex` — sRGB hex code (`#RRGGBB`)
- `ok` — OKLab coordinates (`{L, a, b}`)

Both sRGB (for immediate use) and OKLab (for further computation) are provided.

Config Echo. The effective configuration used, enabling reproducibility. Callers can log, store, or transmit the config to regenerate the same palette later.

Diagnostics. Quality metrics and constraint status for transparency.

10.4 Diagnostic Information

Diagnostics report quality metrics and constraint status:

Metric	Description
minDeltaE	Smallest perceptual distance between adjacent swatches
maxDeltaE	Largest perceptual distance between adjacent swatches
meanDeltaE	Average distance
constraintViolations	Count of constraint violations
gamutCorrections	Count of colors requiring gamut mapping
traversalStrategy	Algorithm used (e.g., “arc-length”)

Diagnostics are informational, not prescriptive. The engine does not fail on constraint violations—it reports them and produces output anyway.

10.5 Scope Boundaries

Design Decision: Engine Core vs. Caller Responsibilities

Choice: The engine core owns palette generation only. Color naming, accessibility checking (WCAG), color blindness simulation, and ICC color management are explicitly caller responsibilities.

Rationale: Keeping the core lean enables production-grade performance (5.6M colors/second) and avoids bloating with features that vary by use case. Language bindings and applications can add these capabilities as needed.

Alternatives Considered:

- *Integrated accessibility checking* — Rejected: varies by standard (WCAG 2.1, 3.0), adds dependencies
- *Built-in color naming* — Rejected: naming systems are culturally/contextually specific
- *Color blindness simulation* — Rejected: simulation algorithms evolve independently

Reference: §11.3, Appendix D

Features that can be implemented by callers using the core output are not built into the engine. Examples of excluded features: “play animation at 30fps”, “apply palette to image”, “export to Photoshop swatch file”.

10.6 Performance Characteristics

The C-core implementation achieves production-grade performance:

Metric	Value
Per-color generation	Microsecond range
Throughput	5.6 million colors/second ¹
Memory	Minimal; no caching required

Computational complexity:

- Journey construction: $O(1)$ per anchor (fixed-degree Bézier)
- Arc-length parameterisation: $O(n)$ precomputation, $O(1)$ lookup
- Sampling: $O(n)$ for n colors with constraint checking

This performance enables real-time palette generation in interactive applications, eliminates the need for precomputation or caching, and supports high-throughput batch processing.

Chapter 11

Caller Responsibilities

11.1 Input Validation

The engine performs basic input validation, but callers should validate inputs before submission:

- **Anchor colors** — Should be valid color specifications. Out-of-gamut anchors are accepted but may require correction during palette generation.
- **Count** — Must be a positive integer. Requests for zero or negative counts return an error.
- **Parameters** — Should respect documented ranges. The engine clamps out-of-range values but callers should validate to avoid surprises.
- **Presets** — Unknown preset names return an error.

11.2 Engine Guarantees

The engine provides the following guarantees:

1. **Output count matches request.** The palette contains exactly the requested number of colors.
2. **All outputs are valid sRGB.** Every color in the palette is displayable on standard monitors.
3. **Anchors appear in output.** When anchors are specified, they appear at their designated positions in the palette.
4. **Perceptual constraints are honored.** Adjacent colors differ by at least $\Delta_{\min}$ (where achievable).

5. **Determinism is maintained.** Identical inputs produce identical outputs across calls and platforms.

11.3 Caller Responsibilities

The engine generates palettes; callers are responsible for context-specific concerns:

Accessibility Testing. The engine does not check WCAG contrast ratios. Callers must verify that text-background pairings meet accessibility requirements for their target compliance level (AA, AAA).

Color Blindness. The engine does not simulate or optimize for color vision deficiencies. Callers concerned with color-blind accessibility should test generated palettes with appropriate simulation tools.

Context Suitability. Print and screen have different requirements (Poynton, [2012](#)). Callers must verify that generated palettes are suitable for their target medium, including:

- Color gamut (sRGB output may not match CMYK print)
- Viewing conditions (ambient light, screen calibration)
- Substrate (paper type, coating)

Application Constraints. Brand colors, corporate guidelines, and domain-specific requirements are caller concerns. The engine provides building blocks; callers assemble them appropriately.

Reproducibility. Callers should store the config echo returned by the engine if they need to regenerate palettes later. The engine does not maintain history.

11.4 Error Handling

The engine handles errors gracefully:

- **Invalid inputs** — Return an error with diagnostic message. No partial output is produced.

- **Constraint conflicts** — Resolved using the priority hierarchy (§4.5). Adaptations are documented in diagnostics.
- **Impossible requests** — The closest valid alternative is returned. For example, requesting 100 visually distinct colors from a narrow hue range will produce fewer distinct steps.
- **No silent failures** — All constraint violations, gamut corrections, and adaptations appear in the diagnostics output.

Chapter 12

Conclusion and Future Directions

12.1 Summary

This specification presents the Color Journey Engine, a deterministic system for generating perceptually-uniform color palettes. The engine addresses a persistent challenge in digital color tools: how to create multi-color palettes that feel coherent while respecting the non-linear, device-dependent nature of human color perception.

The solution rests on three pillars. First, the OKLab color space [Ottosson, 2020](#) provides a perceptually uniform foundation where equal numerical distances correspond to equal perceived differences. Second, the journey metaphor—implemented through cubic Bézier curves [Farin, 2002](#) with arc-length parameterisation—ensures smooth, coherent transitions between colors. Third, the constraint system ($\Delta_{\min}$, $\Delta_{\max}$, JND-based adaptive sampling) guarantees that generated colors are both distinguishable and aesthetically pleasing.

The result is a pure function that transforms configuration into palette: stateless, deterministic, and predictable. For quick navigation during team discussions, see Appendix D for a concept map and decision rationale summaries.

12.2 Key Contributions

This specification makes several contributions to the practice of algorithmic color palette generation:

1. **Formal Specification.** A complete, reproducible specification for palette generation that eliminates the “magic” often present in color tools. Same inputs yield same outputs, always.

2. **Mood Expansion Algorithm.** A novel approach for generating coherent multi-color journeys from a single anchor, using style parameters to expand in perceptually meaningful directions. To our knowledge, this single-anchor expansion approach has not been formally documented elsewhere.
3. **Perceptual Constraint Framework.** A system for ensuring minimum distinguishability while allowing stylistic variation. The specific threshold values ($\Delta_{\min} \approx 2.0$, $\Delta_{\max} \approx 5.0$) are design heuristics informed by color science literature (Fairchild, 2013) and practical experimentation, not empirically-validated constants.
4. **Two-Layer Gamut Management.** A prevention-then-correction strategy following established gamut mapping principles (Morovič, 2008), maintaining journey character while guaranteeing displayable output.
5. **Clear Responsibility Boundaries.** Explicit delineation between engine core and caller responsibilities, enabling a lean, high-performance core while supporting extension.

Limitations and Caveats

Several aspects of this specification warrant further investigation:

- The perceptual weights for velocity calculation (§6.3) are design heuristics, not empirically-validated values.
- OKLab’s JND correspondence ($\Delta E \approx 1.0$) is a design target (Ottosson, 2020); formal perceptual validation specific to OKLab remains limited.
- The constraint thresholds ($\Delta_{\min}$, $\Delta_{\max}$) and subdivision caps represent engineering judgment; alternative values may be appropriate for different use cases.

12.3 Future Directions

Several directions merit future investigation:

Extended Color Spaces. While the current specification targets sRGB output, the OKLab foundation supports wider gamuts. Future versions could target Display P3 or Rec. 2020 while maintaining backward compatibility.

Accessibility Integration. Though accessibility checking remains a caller responsibility, future work could explore optional WCAG-aware generation modes that constrain palettes to meet contrast requirements.

Multi-Palette Harmonisation. Design systems often require multiple coordinated palettes (primary, secondary, semantic). Algorithms for generating harmonised palette families present an interesting extension.

Symmetry-Constrained Optimisation. The current engine constructs journeys through direct path specification. Future work could explore optimisation-based approaches that search for “optimal” journeys maximising perceptual distance while satisfying constraints. Recent work on symmetry-constrained search in combinatorial domains (Moosbauer and Poole, 2025) suggests that constraining search to symmetric or well-behaved curve families could dramatically reduce optimisation complexity—a principle that could apply to journey curve optimisation.

Preset Discovery. The current preset system encodes expert knowledge manually. Machine learning approaches could discover effective presets from successful palette examples.

Real-Time Animation. The engine’s performance enables real-time generation. Formalising smooth interpolation between palettes could support animated color transitions in interactive applications.

The Color Journey Engine provides a foundation for these extensions while delivering immediate value: deterministic, perceptually-sound palette generation with a clean, predictable API.

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Appendix A

Preset Reference

A.1 Overview

Presets encode tested parameter combinations that produce aesthetically coherent results for common use cases. Each preset represents a named configuration that has been validated through practical experimentation and user feedback.

Presets serve three purposes:

1. **Quick Start** — Provide sensible defaults for common scenarios without requiring parameter knowledge
2. **Learning Aid** — Demonstrate effective parameter combinations as examples
3. **Starting Points** — Serve as baselines for customisation; explicit parameters override preset defaults

When a preset is specified in the API call (§10.2), it expands to a parameter set. Any explicit parameters provided alongside the preset will override the preset's defaults, enabling fine-tuning without starting from scratch.

A.2 Preset Table

The following table provides the complete preset reference with all parameter values. Parameters not listed use engine defaults (§10.2).

Table A.1: Preset Reference

Preset	Use Case	Temp	Int	Smooth	Mode	Loop
linear	Simple interpolation	0.0	0.0	0.7	journey	open
smooth	Gentle gradients	0.0	0.3	0.9	journey	open
vivid	High-impact palettes	0.0	0.8	0.6	journey	open
muted	Subdued, subtle	0.0	0.2	0.8	journey	open
sunset	Warm gradients	0.7	0.6	0.8	journey	open
sunrise	Dawn warmth	0.5	0.5	0.9	journey	open
ocean	Cool aquatic tones	−0.6	0.5	0.7	journey	open
forest	Natural greens	−0.3	0.4	0.8	journey	open
autumn	Fall foliage	0.6	0.7	0.6	journey	open
arctic	Ice and snow	−0.8	0.3	0.9	journey	open
neon	Electric, vibrant	0.0	1.0	0.5	journey	closed
pastel	Soft, light	0.0	0.2	0.9	journey	open
earth	Warm naturals	0.4	0.3	0.8	journey	open
monochrome	Single-hue depth	0.0	0.1	0.95	journey	open
categorical	Distinct categories	0.0	0.3	0.5	categorical	open
diverging	Two-pole data	0.0	0.5	0.7	journey	open
sequential	Ordered data	0.0	0.4	0.8	journey	open
breathing	Pulsing animation	0.0	0.4	0.95	journey	pingpong
cycling	Continuous loop	0.0	0.5	0.9	journey	closed
evolving	Progressive shift	0.0	0.5	0.8	journey	phased

Key. **Temp** = Temperature (−1 to +1); **Int** = Intensity (0 to 1); **Smooth** = Smoothness (0 to 1); **Mode** = journey or categorical (§6.1, §6.2); **Loop** = open, closed, pingpong, or phased (§7.1–§7.5).

A.3 Preset Descriptions

A.3.1 Basic Presets

linear The most basic preset—pure linear interpolation between anchors with no curve enhancement. Useful as a baseline for comparison or when mathematical predictability is paramount. Zero intensity means control points lie exactly on the anchor line (§3.5).

smooth Gentle, flowing gradients with high smoothness for seamless C^1 continuity at anchor junctions. Appropriate for backgrounds, ambient effects, and contexts where the color transition should feel natural and unobtrusive.

vivid High-intensity curves that create dramatic chromatic excursions between anchors. The path swings through higher-chroma regions, producing bold, attention-grabbing palettes. Best for creative applications where visual impact is desired.

muted Subdued palettes with low intensity and high smoothness. Produces sophisticated, understated color combinations suitable for professional interfaces, document themes, and contexts requiring visual restraint.

A.3.2 Mood-Based Presets

sunset Warm temperature bias (+0.7) routes the hue path through reds, oranges, and yellows. High smoothness ensures the gradient feels like a natural sky progression. Ideal for conveying warmth, comfort, or evening atmosphere.

sunrise Similar to sunset but slightly cooler and softer, capturing the gentler warmth of dawn. Higher smoothness (0.9) creates an ethereal quality.

ocean Cool temperature bias (−0.6) emphasises blues, cyans, and aquatic greens. The path avoids warm hues entirely, creating cohesive underwater or maritime palettes.

forest Moderate cool bias with natural green emphasis. Lower intensity preserves the organic, muted quality of woodland colors without artificial saturation.

autumn Warm bias with higher intensity to capture the vibrant oranges, reds, and golds of fall foliage. Slightly lower smoothness allows for more distinct color “stops” resembling individual leaves.

arctic Strong cool bias (-0.8) with low intensity and high smoothness for ice, snow, and cold atmospheres. The resulting palettes feel crisp and minimal.

A.3.3 Stylistic Presets

neon Maximum intensity with closed loop for continuous cycling. Creates electric, high-energy palettes suitable for nightclub aesthetics, gaming interfaces, or attention-grabbing animations.

pastel Low intensity with very high smoothness produces soft, desaturated palettes. The colors feel gentle and approachable—suitable for children’s applications, wellness themes, or light UI backgrounds.

earth Warm bias with restrained intensity for natural, organic color combinations. Browns, tans, and warm neutrals dominate, suitable for outdoor, craft, or sustainability themes.

monochrome Minimal intensity (0.1) with near-maximum smoothness for single-hue depth exploration. When combined with a single anchor (§3.3), produces sophisticated tonal variations of one color.

A.3.4 Data Visualisation Presets

categorical Uses Categorical Mode (§6.2) to maximise perceptual distance between all color pairs. Essential for charts, legends, and any context where colors represent distinct categories that must be easily distinguished.

diverging Journey Mode with balanced parameters, intended for two-anchor use where one anchor represents a negative extreme and the other a positive extreme. The neutral midpoint is automatically generated through interpolation.

sequential Optimized for ordered data (low-to-high, early-to-late). Smooth progression with moderate intensity creates clear directionality in the palette.

A.3.5 Animation Presets

breathing Ping-pong loop (§7.3) with very high smoothness for pulse effects. The color smoothly advances then retreats, creating a “breathing” rhythm without any discontinuity at reversal points.

cycling Closed loop (§7.2) for continuous rotation. High smoothness ensures no visible “seam” where the loop wraps from end to beginning.

evolving Phased loop (§7.5) for long-running animations that should feel alive. Each cycle shifts slightly, preventing the static feeling of exact repetition.

A.4 Preset Selection Guide

If you need...	Try preset
Simple, predictable gradient	linear
Warm, inviting atmosphere	sunset or earth
Cool, calm, professional	ocean or arctic
High-energy, attention-grabbing	vivid or neon
Soft, gentle, approachable	pastel or muted
Chart/graph categories	categorical
Heatmap or sequential data	sequential
Looping animation	cycling or breathing
Tonal variations of one color	monochrome

Appendix B

Mathematical Notation

B.1 Color Space Notation

This section provides a complete reference for all mathematical notation used throughout the paper. Symbols are grouped by domain for easy lookup.

B.1.1 OKLab Coordinates

Symbol	Range	Meaning
L	$[0, 1]$	Lightness in OKLab (0 = black, 1 = white)
a	$\approx [-0.4, 0.4]$	Green–red opponent axis (negative = green, positive = red)
b	$\approx [-0.4, 0.4]$	Blue–yellow opponent axis (negative = blue, positive = yellow)
(L, a, b)	—	A color point in OKLab Cartesian coordinates

B.1.2 OKLCh Coordinates (Cylindrical Form)

Symbol	Range	Meaning
L	$[0, 1]$	Lightness (identical to OKLab)
C	$[0, \sim 0.4]$	Chroma: $C = \sqrt{a^2 + b^2}$ (colorfulness/saturation)
h	$[0, 2\pi)$	Hue angle: $h = \text{atan2}(b, a)$ (radians)
(L, C, h)	—	A color point in OKLCh cylindrical coordinates

B.1.3 Conversion Matrices

Symbol	Meaning
M_1	XYZ to LMS transformation matrix (Equation 2.5)
M_2	LMS to OKLab transformation matrix (Equation 2.6)

B.2 Distance and Constraint Notation

B.2.1 Perceptual Distance

Symbol	Definition	Meaning
ΔE_{OK}	$\sqrt{(L_2 - L_1)^2 + (a_2 - a_1)^2 + (b_2 - b_1)^2}$	Perceptual distance in OKLab
ΔE	—	Shorthand for ΔE_{OK} when context is clear
JND	≈ 1.0	Just-noticeable difference threshold
D	—	Total perceptual distance along a segment

B.2.2 Constraint Thresholds

Symbol	Value	Meaning
$\Delta_{\min}$	≈ 2.0	Minimum required distance between adjacent swatches
$\Delta_{\max}$	≈ 5.0	Maximum allowed distance between adjacent swatches
n	$\min(5, \lceil D/\Delta_{\max} \rceil)$	Number of segments for adaptive sampling

B.3 Curve and Path Notation

B.3.1 Bézier Curve Elements

Symbol	Range	Meaning
$B(t)$	—	Bézier curve evaluated at parameter t
$\gamma(t)$	—	Journey path function at parameter t
t	$[0, 1]$	Curve parameter (not arc-length)
s	$[0, S]$	Arc-length parameter (uniform perceptual distance)
S	—	Total arc length of the path
P_0	—	Start control point (equals start anchor)
P_1	—	First interior control point
P_2	—	Second interior control point
P_3	—	End control point (equals end anchor)

B.3.2 Path Construction

Symbol	Meaning
γ_i	The i -th segment of the journey path
$\gamma'(t)$	Tangent vector (derivative) of the path at t
$ \gamma'(t) $	Magnitude of tangent (instantaneous speed)
C^0	Position continuity at junction points
C^1	Tangent (direction) continuity at junction points

B.3.3 Bézier Curve Formula

The cubic Bézier curve (Equation 3.5):

$$\gamma(t) = (1 - t)^3 P_0 + 3(1 - t)^2 t P_1 + 3(1 - t) t^2 P_2 + t^3 P_3$$

Tangent vectors at endpoints:

$$\gamma'(0) = 3(P_1 - P_0)$$

$$\gamma'(1) = 3(P_3 - P_2)$$

B.4 Anchor Notation

Symbol	Range	Meaning
A_i	—	The i -th anchor color
A_1	—	First anchor (journey start)
A_m	—	Last anchor (journey end); $m \leq 5$
m	$[1, 5]$	Number of anchors provided

B.5 Parameter Notation

B.5.1 Primary Style Parameters

Symbol	Range	Default	Meaning
T	$[-1, +1]$	0	Temperature (cool to warm hue bias)
ι	$[0, 1]$	0.5	Intensity (control point offset scale)
σ	$[0, 1]$	0.7	Smoothness (C^1 continuity strength)
N	$\mathbb{Z}^+$	—	Requested palette size (count)

B.5.2 Secondary Style Parameters

Symbol	Range	Default	Meaning
λ	$[-1, +1]$	0	Lightness bias (darker to lighter)
χ	$[0, 2]$	1	Chroma multiplier (desaturate to saturate)
κ	$[0, 2]$	1	Contrast (compress to expand L range)
v	$[0, 2]$	1	Vibrancy (selective chroma boost)

B.5.3 Derived Values

Expression	Meaning
$\vec{s}_1, \vec{s}_2$	Style-determined offset vectors for control points
$\hat{v}_{\text{base}}$	Unit vector in shortest hue direction
$\hat{v}_{\text{warm/cool}}$	Unit vector toward warm or cool hues
α	Blending coefficient for temperature mixing
C_{max}	Maximum chroma at a given (L, h) within gamut

B.6 Variation and Determinism Notation

Symbol	Range	Meaning
$\delta_L, \delta_a, \delta_b$	varies	Perturbation offsets in OKLab dimensions
seed	$\mathbb{Z}$	PRNG seed for deterministic variation
k	$\mathbb{Z}^+$	Cycle index for phased loops

B.7 Loop Strategy Notation

Symbol	Range	Meaning
$J(t)$	—	Journey function at parameter t
$J_k(t)$	—	Journey on cycle k (phased loops)
u	$[0, 2]$	Extended parameter for ping-pong
$\tilde{t}$	$[0, 1]$	Remapped parameter after ping-pong transform

B.7.1 Loop Parameter Transforms

Ping-pong transform:

$$\tilde{t} = \begin{cases} u & 0 \leq u < 1 \\ 2 - u & 1 \leq u < 2 \end{cases}$$

Möbius twist (chromatic inversion at midpoint):

$$J_{\text{mobius}}(1) = (L, -a, -b) \quad \text{when } J_{\text{mobius}}(0) = (L, a, b)$$

Phased shift:

$$J_k(t) = J_0(t) + k \cdot \text{shift}$$

B.8 Gamut Notation

Symbol	Meaning
G_{sRGB}	The sRGB gamut volume mapped into OKLab space
$\gamma(t) \in G_{\text{sRGB}}$	Constraint that all path points are displayable
C'	Corrected (reduced) chroma after gamut mapping
L'	Adjusted lightness (if gamut mapping required)

B.9 Perceptual Velocity Notation

Symbol	Typical Value	Meaning
v	—	Perceptual velocity (weighted rate of change)
w_L	1.0	Weight for lightness change rate
w_C	~ 1.2	Weight for chroma change rate
w_h	$\sim 1.5\text{--}2.0$	Weight for hue change rate
$\frac{dL}{dt}, \frac{dC}{dt}, \frac{dh}{dt}$	—	Rates of change in each dimension

Appendix C

Implementation Examples

C.1 Basic Usage

This appendix provides practical code examples demonstrating common usage patterns for the Color Journey Engine. All examples use a JavaScript-like pseudocode that maps directly to the API structure defined in §10.2.

C.1.1 Simple Two-Anchor Gradient

The most basic usage: interpolate between two colors to create a smooth gradient.

```
const palette = generatePalette({
  count: 5,
  anchors: [ '#FF6B6B', '#4ECDC4' ]
});

// Result: 5 colors from coral to teal
// [
//   { hex: '#FF6B6B', ok: { L: 0.70, a: 0.14, b: 0.05 } },
//   { hex: '#E08B7A', ok: { L: 0.68, a: 0.08, b: 0.03 } },
//   { hex: '#8FB8A9', ok: { L: 0.72, a: -0.03, b: 0.02 } },
//   { hex: '#5DC5B8', ok: { L: 0.74, a: -0.08, b: 0.01 } },
//   { hex: '#4ECDC4', ok: { L: 0.76, a: -0.10, b: 0.00 } }
// ]
```

Listing C.1: Basic Two-Anchor Gradient

C.1.2 Multi-Anchor Journey

Three or more anchors create a journey that passes through each waypoint.

```
const palette = generatePalette({
  count: 9,
  anchors: ['#1A1A2E', '#E94560', '#0F3460'],
  smoothness: 0.8
});

// Result: dark blue -> vibrant red -> deep blue
// Anchors appear at positions 0, 4, and 8 (approximately)
// Intermediate colors flow smoothly between them
```

Listing C.2: Three-Anchor Journey

C.2 Single-Anchor Expansion

Single-anchor mode (§3.3) generates harmonious variations from one color, useful for UI state variations and monochromatic palettes.

```
const palette = generatePalette({
  count: 7,
  anchors: ['#6B5B95'], // Single purple anchor
  preset: 'monochrome'
});

// Engine expands along lightness and subtle hue directions
// Result: tonal variations centred on the anchor
// Includes lighter, darker, and slightly hue-shifted variants
```

Listing C.3: Single-Anchor Mood Expansion

C.2.1 Brand Color State Variants

Generate hover, active, and disabled states from a single brand color.

```
const brandColor = '#2563EB'; // Primary blue

const states = generatePalette({
  count: 5,
  anchors: [brandColor],
  intensity: 0.2, // Subtle variations
  smoothness: 0.95 // Very smooth transitions
});
```

```
// Use positions for different states:
// states[0] - darkest (pressed/active)
// states[2] - anchor (normal state)
// states[4] - lightest (hover/highlight)
```

Listing C.4: UI State Variations

C.3 Categorical Mode

Categorical Mode (§6.2) maximises perceptual distance between all colors, essential for data visualisation.

```
const chartColors = generatePalette({
  count: 6,
  anchors: ['#2C3E50'], // Starting point hint
  mode: 'categorical'
});

// Result: 6 maximally distinct colors
// Each pair has maximum perceptual separation
// Suitable for pie charts, bar charts, legends
```

Listing C.5: Categorical Palette for Chart

C.3.1 Categorical with Fixed Anchors

Pin specific semantic colors while filling remaining slots.

```
const statusColors = generatePalette({
  count: 5,
  anchors: [
    '#DC2626', // Red (error) - fixed
    '#16A34A', // Green (success) - fixed
    '#2563EB'  // Blue (info) - fixed
  ],
  mode: 'categorical'
});

// Anchors preserved exactly
// Two additional colors generated with maximum distinction
```

Listing C.6: Semantic Categorical Palette

C.4 Temperature Control

Temperature (§5.1) biases the hue path toward warm or cool regions.

```
// Same anchors, different temperatures
const coolPalette = generatePalette({
  count: 7,
  anchors: ['#8B5CF6', '#10B981'], // Purple to green
  temperature: -0.6 // Cool: via blues
});

const warmPalette = generatePalette({
  count: 7,
  anchors: ['#8B5CF6', '#10B981'], // Same anchors
  temperature: +0.6 // Warm: via reds/yellows
});

// coolPalette passes through cyans and teals
// warmPalette passes through magentas and yellows
// Dramatically different character, same endpoints
```

Listing C.7: Warm Temperature Bias

C.5 Loop Strategies

Loop strategies (§7.1–§7.5) control how the palette wraps or cycles.

C.5.1 Closed Loop for Animation

```
const cyclingPalette = generatePalette({
  count: 12,
  anchors: ['#E74C3C', '#3498DB', '#2ECC71'],
  loop: 'closed',
  smoothness: 0.9
});

// Last color transitions smoothly to first
```



```
// Output: 12 colors, no duplicate at end
// Loop externally: [...cyclingPalette, cyclingPalette[0], ...]
// Seamless continuous animation
```

Listing C.8: Closed Loop for Continuous Cycling

C.5.2 Ping-Pong for Breathing Effect

```
const breathingPalette = generatePalette({
  count: 16,
  anchors: ['#1E3A5F', '#60A5FA'], // Dark to light blue
  loop: 'pingpong',
  preset: 'breathing'
});

// Smooth oscillation: dark -> light -> dark
// 16 colors cover full forward-backward cycle
// No discontinuity at reversal point
```

Listing C.9: Ping-Pong Breathing Animation

C.5.3 Phased Loop for Evolution

```
const evolvingPalette = generatePalette({
  count: 24,
  anchors: ['#F59E0B', '#8B5CF6'],
  loop: 'phased'
  // Each cycle shifts hue slightly
});

// Cycle 1: Amber -> Violet
// Cycle 2: Amber+shift -> Violet+shift
// Never exactly repeats, feels alive
```

Listing C.10: Phased Loop for Progressive Evolution

C.6 Full Configuration

Comprehensive example demonstrating all parameters.

```
const result = generatePalette({
  // Required
  count: 8,
  anchors: ['#1A1A2E', '#E94560', '#0F3460'],

  // Primary style controls
  temperature: 0.2,      // Slightly warm bias
  intensity: 0.7,        // Dramatic curves
  smoothness: 0.8,       // Smooth anchor transitions

  // Secondary controls
  lightness: 0.0,        // No lightness shift
  chroma: 1.2,           // Boost saturation 20%
  contrast: 1.1,         // Slightly increase contrast
  vibrancy: 1.0,         // Default vibrancy

  // Behavior
  mode: 'journey',       // Smooth sequential path
  loop: 'open',          // Start to finish, no wrap

  // Reproducibility
  seed: 42               // Deterministic output
});

// Access output components
console.log(result.palette); // Array of 8 color objects
console.log(result.config);  // Effective configuration used
console.log(result.diagnostics); // Quality metrics
```

Listing C.11: Complete Configuration Example

C.7 Working with Output

C.7.1 Accessing Color Formats

```
const result = generatePalette({
  count: 5,
  anchors: ['#FF6B6B', '#4ECDC4']
});
```

```
// Each swatch has both formats
result.palette.forEach((swatch, i) => {
  console.log('Color ${i}:');
  console.log('  Hex: ${swatch.hex}');           // For CSS/
    display
  console.log('  OKLab L: ${swatch.ok.L}');       // For
    computation
  console.log('  OKLab a: ${swatch.ok.a}');
  console.log('  OKLab b: ${swatch.ok.b}');
});
```

Listing C.12: Using Output Color Formats

C.7.2 Using Diagnostics

```
const result = generatePalette({
  count: 20,
  anchors: ['#FFFFFF', '#000000'], // White to black
  chroma: 2.0                       // High chroma request
});

const diag = result.diagnostics;
console.log('Min Delta E: ${diag.minDeltaE}');
console.log('Max Delta E: ${diag.maxDeltaE}');
console.log('Mean Delta E: ${diag.meanDeltaE}');
console.log('Gamut corrections: ${diag.gamutCorrections}');

if (diag.gamutCorrections > 0) {
  console.log('Some colors were desaturated to fit sRGB gamut');
}

if (diag.constraintViolations > 0) {
  console.log('Some step sizes exceeded ideal range');
}
```

Listing C.13: Checking Quality Diagnostics

C.8 Common Patterns

C.8.1 Diverging Color Scale

```
// Diverging: negative (red) -> neutral -> positive (blue)
const divergingScale = generatePalette({
  count: 11,
  anchors: ['#DC2626', '#F5F5F5', '#2563EB'],
  preset: 'diverging'
});

// Index 5 is neutral (near white)
// Indices 0-4 are increasingly red
// Indices 6-10 are increasingly blue
```

Listing C.14: Diverging Scale for Data

C.8.2 Sequential Scale with Preset

```
const heatmapScale = generatePalette({
  count: 9,
  anchors: ['#FEF3C7', '#DC2626'], // Light yellow to red
  preset: 'sequential'
});

// Low values: light warm colors
// High values: intense red
// Ideal for heatmaps, intensity visualisations
```

Listing C.15: Sequential Scale Using Preset

C.8.3 Reproducing a Palette

```
// Generate and store config
const result = generatePalette({
  count: 8,
  anchors: ['#FF6B6B', '#4ECDC4'],
  temperature: 0.3,
  seed: 12345
});

// Store the effective config (compact)
const storedConfig = result.config;
localStorage.setItem('myPalette', JSON.stringify(storedConfig));
```

```
// Later: regenerate identical palette
const savedConfig = JSON.parse(localStorage.getItem('myPalette'))
;
const regenerated = generatePalette(savedConfig);

// regenerated.palette is identical to original
```

Listing C.16: Storing and Reproducing Configuration

Appendix D

Quick Reference

D.1 Quick Reference for Team Discussions

This appendix provides rapid lookup of key concepts, design decisions, and section references for use during code reviews and design discussions.

D.1.1 Concept Map

D.1.2 Decision Rationale Summary

This section summarises the key design decisions documented throughout the paper. Each decision box in the main text explains the choice, rationale, and rejected alternatives.

- **Mood expansion** (§3.3): Single-anchor path uses lightness-weighted directions; naive hue spin rejected to avoid non-perceptual artifacts.
- **Loop output** (§7.6): Unrolled arrays prevent nested structures; segmented outputs rejected for caller simplicity.
- **Gamut handling** (§8.2–§8.3): Two-layer approach (prevention + correction); hard clipping rejected to preserve hue; HSV fallback rejected to maintain perceptual uniformity.
- **API scope** (§10.5): Engine owns core generation; naming, accessibility, and color blindness simulation are caller responsibilities.

Topic	Section	Key Insight
Why OKLab?	§2.4	Perceptual uniformity; CAM16-level accuracy, low cost
OKLCh vs OKLab	§2.4.6	Cylindrical for hue ops; Cartesian for distance
Anchor guarantee	§3.2	Anchors appear exactly, not “close to”
Mood expansion	§3.3	Single-anchor uses lightness-weighted directions
Bézier curves	§3.5	Cubic curves for flexibility and C^1 continuity
Arc-length sampling	§3.6	Uniform perceptual steps, not parametric steps
JND threshold	§4.1	~ 1.0 theoretical, ~ 2.0 practical
$\Delta_{\min}$ constraint	§4.2	JND-based ($\sim 2 \Delta E$); ensures distinguishability
$\Delta_{\max}$ constraint	§4.3	Coherence cap ($\sim 5 \Delta E$); no jarring jumps
Adaptive sampling	§4.4	Auto-subdivide long segments, cap at 5
Temperature control	§5.1	Bias hue path warm (+) or cool (−)
Intensity control	§5.2	Scale control point offset from anchor line
Smoothness control	§5.3	C^1 continuity strength at junctions
Journey vs Categorical	§6.4	Sequential flow vs maximum pairwise distinction
Perceptual velocity	§6.3	Hue changes feel “faster” than L/C changes
Loop output	§7.6	Flat array (unrolled), no nested structure
Gamut prevention	§8.2	Control points prefer moderate chroma
Gamut correction	§8.3	Reduce chroma, preserve hue (never shift hue)
Determinism	§9.1	Same inputs = same outputs (pure function)
Variation layer	§9.2	Seeded PRNG; anchors never perturbed
API philosophy	§10.1	Stateless, focused, no side effects
Performance	§10.6	Microsecond-range; 5.6M colors/second

Table D.1: Quick reference mapping topics to sections

D.1.3 Parameter Quick Reference

D.1.4 Output Structure Quick Reference

D.1.5 Common Use Cases

D.1.6 Perceptual Distance Interpretation

D.1.7 Troubleshooting Guide

D.1.8 Constraint Summary

Parameter	Range	Default	Effect
count	$\mathbb{Z}^+$	(required)	Number of output colors
anchors	1–5 colors	(required)	Key colors to pass through
temperature	$[-1, +1]$	0	Warm/cool hue bias
intensity	$[0, 1]$	0.5	Curve drama (control point offset)
smoothness	$[0, 1]$	0.7	C^1 continuity at anchors
lightness	$[-1, +1]$	0	Global lightness shift
chroma	$[0, 2]$	1	Saturation multiplier
contrast	$[0, 2]$	1	Lightness range expansion
vibrancy	$[0, 2]$	1	Selective chroma boost
mode	enum	journey	journey or categorical
loop	enum	open	open, closed, pingpong, etc.
seed	integer	null	Variation seed (omit = no variation)
preset	string	null	Named parameter set

Table D.2: Input parameters quick reference

Field	Contents
palette[]	Array of swatch objects
palette[i].hex	sRGB hex code (#RRGGBB)
palette[i].ok	OKLab coordinates {L, a, b}
config	Effective configuration (for reproducibility)
diagnostics.minDeltaE	Smallest adjacent distance
diagnostics.maxDeltaE	Largest adjacent distance
diagnostics.meanDeltaE	Average distance
diagnostics.gamutCorrections	Count of gamut-mapped colors
diagnostics.constraintViolations	Count of constraint issues

Table D.3: Output structure quick reference

Use Case	Mode	Recommended Preset
Smooth gradient back-ground	journey	smooth
Warm sunset atmosphere	journey	sunset
Cool professional theme	journey	ocean or arctic
Chart legend colors	categorical	categorical
Heatmap scale	journey	sequential
Positive/negative diverging	journey	diverging
UI state variations	journey	monochrome
Continuous animation loop	journey (closed)	cycling
Pulsing/breathing effect	journey (pingpong)	breathing
High-energy vibrant palette	journey	neon or vivid
Soft, approachable colors	journey	pastel

Table D.4: Use case to preset mapping

ΔE Range	Interpretation
< 1.0	Imperceptible to most observers
1.0 – 2.0	Barely perceptible, subtle
2.0 – 3.0	Noticeable, clear difference
3.0 – 5.0	Obvious difference, still smooth
> 5.0	Pronounced difference, may feel like a “step”

Table D.5: Perceptual distance interpretation guide (from §4.1)

Symptom	Likely Cause / Solution
Colors look washed out	High chroma caused gamut mapping; reduce chroma
Unexpected color in middle	Temperature bias routing through different hues; check temperature
Visible “jump” between colors	Step size exceeds $\Delta_{\max}$; add anchors or increase count
Colors too similar	Anchors too close; use different anchors or Categorical Mode
Animation has visible seam	Using open loop; switch to closed
Different output each run	Missing seed with variation; provide explicit seed
Palette doesn’t match saved	Engine version changed; store full config for reproducibility

Table D.6: Common issues and solutions

Constraint	Value	Purpose
Maximum anchors	5	Cognitive tractability (§3.2)
$\Delta_{\min}$	$\sim 2.0 \Delta E$	Ensure distinguishability (§4.2)
$\Delta_{\max}$	$\sim 5.0 \Delta E$	Prevent jarring jumps (§4.3)
Max subdivisions	5	Prevent runaway palette growth (§4.3)

Table D.7: Engine constraints summary

Appendix E

OKLab Reference Code

Appendix E: OKLab Reference Implementation (C++ Code)

Source: Björn Ottosson, OKLab, MIT License. See <https://bottosson.github.io/posts/oklab/>

```
struct Lab {float L; float a; float b;};
struct RGB {float r; float g; float b;};

Lab linear_srgb_to_oklab(RGB c)
{
    float l = 0.4122214708f * c.r + 0.5363325363f * c.g + 0.0514459929f * c.b;
    float m = 0.2119034982f * c.r + 0.6806995451f * c.g + 0.1073969566f * c.b;
    float s = 0.0883024619f * c.r + 0.2817188376f * c.g + 0.6299787005f * c.b;

    float l_ = cbrtf(l);
    float m_ = cbrtf(m);
    float s_ = cbrtf(s);

    return {
        0.2104542553f*l_ + 0.7936177850f*m_ - 0.0040720468f*s_,
        1.9779984951f*l_ - 2.4285922050f*m_ + 0.4505937099f*s_,
        0.0259040371f*l_ + 0.7827717662f*m_ - 0.8086757660f*s_,
    };
}

RGB oklab_to_linear_srgb(Lab c)
{
    float l_ = c.L + 0.3963377774f * c.a + 0.2158037573f * c.b;
```

```
float m_ = c.L - 0.1055613458f * c.a - 0.0638541728f * c.b;
float s_ = c.L - 0.0894841775f * c.a - 1.2914855480f * c.b;

float l = l_*l_*l_;
float m = m_*m_*m_;
float s = s_*s_*s_;

return {
    +4.0767416621f * l - 3.3077115913f * m + 0.2309699292f * s,
    -1.2684380046f * l + 2.6097574011f * m - 0.3413193965f * s,
    -0.0041960863f * l - 0.7034186147f * m + 1.7076147010f * s,
};
}
```

License: MIT License. The matrices were updated 2021-01-25 for higher precision and D65 matching. For application-specific needs, you may substitute your own sRGB matrices as appropriate.